



fitipower integrated technology Inc.

JD9365D

User Guide

800RGB x 1280 dot, 16.7M color,
without internal GRAM, a-Si TFT LCD Single Chip Driver

Version 0.01
2017/12/21

INDEX LIST:

1.	REVISION HISTORY	6
2.	JADARD COMMAND LIST	7
2.1.	ALL PAGE.....	7
2.2.	JADARD PAGE 0 COMMAND.....	8
2.3.	JADARD PAGE 1 COMMAND.....	9
2.4.	ALL PAGE COMMAND DESCRIPTION.....	15
2.4.1.	SET_PAGE(ALL Page-RE0h)	15
2.4.2.	SET_PASSWORD(ALL Page-RE1h~RE3h).....	16
2.4.3.	AUTO_DISP_SETTING(ALL Page -RE4h~RE5h).....	17
2.4.4.	SET_WD: Setup watch dog (ALL Page -RE6h~RE8h).....	18
2.4.5.	OTP PROG: OTP Program (ALL Page -RE9h~REFh)	19
2.4.6.	SELF_DIAG(ALL Page -RF0h).....	21
2.5.	JADARD PAGE 0 COMMAND DESCRIPTION	22
2.5.1.	RDCUSTMID: Read Customer ID (Page0-R1Bh)	22
2.5.2.	SEQUENCE CTRL: Power on sequence control (Page0-R70h~R76h)	23
2.5.3.	SET ID: SETID(OTP*5) (Page0-R77h~R7Ch)	26
2.5.4.	SETDSISETUP: Set DSI Setup (Page0-R80h~R82h)	27
2.5.5.	SETDSISETUP: Set DSI Setup2 (Page0-R83h)	29
2.5.6.	UP_START_CTRL (Page0-R84h~R85h).....	30
2.5.7.	ALS: SET_ALS (Page0-R88h~R8Bh).....	31
2.6.	JADARD PAGE 1 COMMAND DESCRIPTION	33
2.6.1.	VCOM_SET: Set VCOM Voltage (Page1-R00h~R02h)	33
2.6.2.	VCOM_R_SET: Set VCOM Voltage when reverse scan (Page1-R03h~R05h)	34
2.6.3.	PWRIC SET: External Power IC timing setting (Page1-R0Ah~R0Eh).....	35
2.6.4.	GAMMA_SET: Set Gama Reference Voltage(Page1-R17h~R1Ch).....	37
2.6.5.	OTP_SET(Page1-R1Dh~R1Eh).....	39
2.6.6.	GATE_POWER_SET: Set GIP Output High Level (Page1-R1Fh~R22h).....	40
2.6.7.	DCDC_CTRL(Page1-R23h~R24h)	42
2.6.8.	POWAMP_CTRL: Set Power Function Related Setting (Page1-R25h~R27h).....	44
2.6.9.	SETSTBA: Set Source Option (Page1- R35h~R36h)	46
2.6.10.	SETPANEL: Set Panel Related (Page1-R37h).....	47
2.6.11.	SETRGBCYC: Set Display Waveform Cycles of RGB Mode (Page1- R38h~R3Fh)	48
2.6.12.	SET TCON: Timing control setting (Page1-R40h~R4Ch).....	51

JD9365D

2.6.13.	SET TE PAD: TE pad function setting (Page1-R4Dh~R53h)	56
2.6.14.	POWER_STATE: Power status setting (Page1-R54h).....	58
2.6.15.	DCDC_SEL: Power Mode and Charge Pump Setting (Page1-R55h~ R5Ch)	59
2.6.16.	SET_GAMMA: Set Gamma output voltage (Page1-R5Dh~ R82h).....	62
2.6.17.	SET DDB: Set Device Descriptor Block (Page1-R83h~R87h).....	64
2.6.18.	SETECO(Page1-R88h~ R8Ch)	65
3.	OTP PROGRAMMING.....	70
3.1.	OTP TABLE.....	70
3.2.	OTP PROGRAMMING FLOW	81
3.3.	PROGRAMMING SEQUENCE	82
3.4.	OTP PROGRAMMING EXAMPLE OF VCOM SETTING.....	83
3.5.	OTP READ EXAMPLE	85



JD9365D

1. Revision History

Version	Date	Description of modification
0.01	2017/12/21	1. New setup

fitipower confidential

2. Jadard Command List

2.1. ALL page

Index	Operation	R/W	D7	D6	D5	D4	D3	D2	D1	D0	Default
	Code										(Hex)
E0	SET_PAGE	R/W	-	-	-	-	-	PAGE[2:0]			00
E1	SET_PASSWD	R/W	PASSWD1[7:0]								00
E2		R/W	PASSWD2[7:0]								00
E3		R/W	PASSWD3[7:0]								00
E4	AUTO_DISP_SETTING	R/W	-	-	-	-	GON	DTE	DISP[1:0]		08
E5		R/W	-	-	DISP_SW_OPT[1:0]		-	BLK_DISABLE	BLK_MODE[1:0]		01
E6	SET_WD	R/W	-	-	-	-	-	-	WD_MODE[1:0]		00
E7		R/W	-	-	-	WD_off	WD_Timer[3:0]			10	
E8		R	-	-	-	-	-	-	WD_FMS	WD_FBLK	00
E9	OTP_PROG	R/W	OTP_MASK[7:0]								00
EA		R/W	-	-	-	-	-	-	OTP_INDEX[9:8]		00
EB		R/W	OTP_INDEX[7:0]								00
EC		R/W	-	-	-	OTP_READ_SINGLE	OTP_WRITE_SINGLE	OTP_LOAD_DISABLE	OTP_INTERRUPT_VPP	OTP_AUTO_PROG	00
ED		R	OTP_RDATA[7:0]								-
EE		R/W	OTP_WDATA[7:0]								00
EF		R/W	OTP_PROG	OTP_VPP_SEL	OTP_PRD	OTP_PWE	-	-	OTP_PTM[1:0]		00

TABLE 2.1: JADARD ALL PAGE COMMAND LIST

JD9365D

2.2. Jadard page 0 command

Index	Operation	R/W	D7	D6	D5	D4	D3	D2	D1	D0	Default
	Code										(Hex)
1B	RDCUSTM_ID	R	CUSTM1[7:0]								FF
		R	CUSTM2[7:0]								FF
		R	CUSTM3[7:0]								FF
		R	CUSTM4[7:0]								FF
		R	CUSTM5[7:0]								FF
		R	CUSTM6[7:0]								FF
		R	CUSTM7[7:0]								FF
		R	CUSTM8[7:0]								FF
70	SEQUENCE_CTRL	R/W	-	DC0H[2:0]			-	DC1H[2:0]			01
71		R/W	-	DC2H[2:0]			-	DC3H[2:0]			23
72		R/W	-	-	-	-	-	DC7H[2:0]			06
73		R/W	DC0L	DC1L	DC2L	DC3L	DC4L	DC5L	DC6L	DC7L	88
74		R/W	T0P[1:0]		T1P[1:0]		T8P[1:0]		-	-	64
75		R/W	-	SOFT0H[2:0]			-	SOFT0L[2:0]			06
76		R	-	-	-	-	-	-	-	POW_ON_STATE	00
77	SETID(OTP*5)	R/W	-	-	-	-	-	-	-	-	00
78		R/W	ID1[7:0]								00
79		R/W	ID2[7:0]								00
7A		R/W	ID3[7:0]								00
7B		R/W	ID4[7:0]								00
7C		R	-	-	-	-	-	ID_OTP_TIMES[2:0]			00
80	SETDSISE_TUP	R/W	-	DSI_INIT0[6:0]							23
81		R/W	DSI_INIT1[7:0]								28
82		R/W	DSI_INIT2[7:0]								07
83	SETDSISE_TUP2	R/W	-	DSI_INIT3[6:0]							0F
84	UP_START_CTRL	R/W	UP_KEY1[7:0]								00
85		R/W	UP_KEY2[7:0]								00
88	SET_ALS	R/W	ALS[31:24]								00
89		R/W	ALS[23:16]								00
8A		R/W	ALS[15:8]								00
8B		R/W	ALS[7:0]								00

TABLE 2.2: JADARD PAGE 0 COMMAND LIST

JD9365D

2.3. Jadard page 1 command

Index	Operation	R/W	D7	D6	D5	D4	D3	D2	D1	D0	Default	
	Code										(Hex)	
00	VCOM_SET(OTP*5)	R/W	-	-	-	-	-	-	-	VCOM_S[8]	00	
01		R/W	VCOM_S[7:0]									32
02		R	-	-	-	-	-	VCOM_OTP_TIMES[2:0]			00	
03	VCOM_R_SET(OTP*5)	R/W	-	-	-	-	-	-	-	VCOM_R_S[8]	00	
04		R/W	VCOM_R_S[7:0]									32
05		R	-	-	-	-	-	VCOM_R_OTP_TIMES[2:0]			00	
06	PFM_CTRL	R/W	-	-	-	-	PFM_CLK[3:0]			01		
07		R/W	DUTYPS[3:0]				DUTYNS[3:0]			11		
08		R/W	DUTYP[3:0]				DUTYN[3:0]			44		
09		R/W		PFM_PDS[2:0]				PFM_PD[2:0]		44		
0A	PWRIC_SET	R/W	-	-	JD_RELD	VCSW_SWAP	EN_VREF_S[3:0]			08		
0B		R/W	PWRIC_SET[7:0]									00
0C		R/W	-	PWRIC_CLK1[2:0]			-	PWRIC_IDLE_CLK1[2:0]			54	
0D		R/W	HX_SOFT	PWRIC_CLK2[2:0]			PWRIC_HXCLK[3:0]			E3		
0E		R/W	-	-	-	-	-	-	LED_PWR_S[1:0]		00	
17	GAMMA_SET	R/W	-	-	-	VGMP_S[8]	-	-	-	VGSP_S[8]	00	
18		R/W	VGMP_S[7:0]									BF
19		R/W	VGSP_S[7:0]									11
1A		R/W	-	-	-	VGMPN_S[8]	-	-	-	VGSN_S[8]	00	
1B		R/W	VGMPN_S[7:0]									BF
1C		R/W	VGSN_S[7:0]									11
1D	OTP_SET	R/W	-	-	VPP_DT[1:0]		-	VPP_VGH_RT[2:0]			01	
1E		R/W	-	VPP_VGH_S[6:0]								03
1F	GATE_POWER	R/W	-	VGH_REG_S[6:0]								34
20		R/W	-	-	VGL_REG_S[5:0]						23	
21		R/W	-	-	VGL_REG2_S[5:0]						19	
22		R/W	-	VGH_REG_SHT	VGL_REG_SHT	VGL_REG2_SHT	VGH_REG_EN	VGL_REG_EN	VGL_REG2_EN	SHT_VGL_REG	0E	

JD9365D

23	DCDC_CTRL	R/W	-	-	-	-	VGH_NC	VGL_NC	VCLREG_OP_CEN	VCL_NC	00
24		R/W	VGHEN	VGLEN	AVDDEN	AVEEEN	VCLN	VDDAEN	VCLREG_EN	VCL2VDDN	FC
25	POWAMP_CTRL	R/W	-	AP[2:0]			-	-	-	-	30
26		R/W	GAS_EN	VCOM_EN	GAS_VCOM_EN	VGM_EN	GAS_AVDD_EN	GAS_AVEE_EN	EN_VDD_VCI	EN_VDDD_I_OVCC	FF
27		R/W	UP_VDD_1P8_EN	VDDD_S1[2:0]			-	VDDD_S2[2:0]			33
35	SETSTBA	R/W	-	GAP[2:0]			SAP[3:0]				24
36		R/W	SAP_Temp[3:0]					SAP2[3:0]			53
37	SETPANEL	R/W	-	Z_line	ENZ[1:0]		SS_PANEL	GS_PANEL	REV_PANEL	BGR_PANEL	00
38	SETRGBCYC	R/W	-	interlace_line	-	-	-	RGB_JDT[2:0]			04
39		R/W	RGB_N_EQ1[7:0]								0C
3A		R/W	RGB_N_EQ2[7:0]								18
3B		R/W	RGB_N_EQ2_Temp[7:0]								18
3C		R/W	RGB_N_EQ3[7:0]								80
3D		R/W	RGB_CHGEN_ON[7:0]								0C
3E		R/W	RGB_CHGEN_OFF[7:0]								80
3F		R/W	RGB_CHGEN_OFF2[7:0]								4E

JD9365D

40	SET_TCON	R/W	-	IP750_1	LN[1:0]	IP750_0	RSO[2:0]			04	
41		R/W	LN[9:2]							A0	
42		R/W	SLT[7:0]							81	
43		R/W	VFP[7:0]							0E	
44		R/W	VBP[7:0]							0A	
45		R/W	HBP[7:0]							16	
46		R/W	TE_Delay[10:3]							00	
47		R/W	BlkinF_num[7:0]							00	
48		R/W	BlkinF_STR[7:0]							00	
49		R/W	BlkinFrame	BlkinF_hsync	-	-	-	TE_Delay[2:0]			40
4A	R/W	FRAME_NUM[1:0]		Auto_Test_en	TEST_PATTERN[3:0]	TEST_PATTERN[3:0]				80	
4B	R/W	TCON_OPT1[15:8]							00		
4C	R/W	TCON_OPT1[7:0]							00		
4D	SET_TE_PAD	R/W	-	-	-	-	-	-	-	00	
4E		R/W	-	-	-	-	TP_VS_EN	TP_VS_WIDTH[10:8]			0D
4F		R/W	TP_VS_WIDTH[7:0]							00	
50		R/W	TP_VS_SHIFT[7:0]							14	
51		R/W	-	-	-	-	-	-	IP_HS_BLK_EN	TP_HS_DISABLE	01
52		R/W	TP_HS_ON[7:0]							48	
53		R/W	TP_HS_OFF[7:0]							70	
54	POWER_STATE	R/W	-	-	-	-	-	-	-	DSTBY	00
55	DCDC_SEL	R/W	-	-	-	-	DCDCM[3:0]			05	
56		R/W	-	-	-	-	AUTO_RT	-	AVDD_RT[1:0]		01
57		R/W	VGH_RT[2:0]			VGL_RT[2:0]			VCL_RT[1:0]		29
58		R/W	-	-	-	AVDD_S[4:0]					0A
59		R/W	VCL_S[2:0]			AVEE_S[4:0]					0A
5A		R/W	-	VGH_S[6:0]							28
5B		R/W	-	-	VGL_S[5:0]						19

JD9365D

5C		R/W	-	-	VDDA_CLK[1:0]	VCL_CLK[1:0]	VGHVGL_CLK[1:0]	15
5D	SET_GAMMA	R/W	-	RPA18[6:0]				7C
5E		R/W	-	RPA17[6:0]				6D
5F		R/W	-	RPA16[6:0]				63
60		R/W	-	RPA15[6:0]				59
61		R/W	-	RPA14[6:0]				57
62		R/W	-	RPA13[6:0]				4A
63		R/W	-	RPA12[6:0]				51
64		R/W	-	RPA11[6:0]				3A
65		R/W	-	RPA10[6:0]				55
66		R/W	-	RPA9[6:0]				53
67		R/W	-	RPA8[6:0]				55
68		R/W	-	RPA7[6:0]				7A
69		R/W	-	RPA6[6:0]				6F
6A		R/W	-	RPA5[6:0]				7F
6B		R/W	-	RPA4[6:0]				75
6C		R/W	-	RPA3[6:0]				72
6D		R/W	-	RPA2[6:0]				62
6E		R/W	-	RPA1[6:0]				2D
6F		R/W	-	RPA0[6:0]				06
70		R/W	-	RNA18[6:0]				7C
71		R/W	-	RNA17[6:0]				6D
72		R/W	-	RNA16[6:0]				63
73		R/W	-	RNA15[6:0]				59
74		R/W	-	RNA14[6:0]				57
75		R/W	-	RNA13[6:0]				4A
76		R/W	-	RNA12[6:0]				51
77		R/W	-	RNA11[6:0]				3A
78		R/W	-	RNA10[6:0]				55
79		R/W	-	RNA9[6:0]				53
7A		R/W	-	RNA8[6:0]				55
7B		R/W	-	RNA7[6:0]				7A
7C		R/W	-	RNA6[6:0]				6F
7D		R/W	-	RNA5[6:0]				7F
7E		R/W	-	RNA4[6:0]				75

JD9365D

7F		R/W	-	RNA3[6:0]							72
80		R/W	-	RNA2[6:0]							62
81		R/W	-	RNA1[6:0]							2D
82		R/W	-	RNA0[6:0]							06
83	SETDDB	R/W	-	-	-	-	-	-	-	00	
84		R/W	DDB1[7:0]							00	
85		R/W	DDB2[7:0]							00	
86		R/W	DDB3[7:0]							00	
87		R/W	DDB4[7:0]							00	
88	SETECO	R/W	-	-	-	-	-	-	-	00	
89		R/W	ECO0[7:0]							03	
8A		R/W	ECO1[7:0]							00	
8B		R/W	ECO2[7:0]							00	
8C		R/W	ECO3[7:0]							00	
93	DCS_CMD_OPT	R/W	-	-	-	-	-	-	NOKIA_CMD_EN	00	
94		R/W	-	-	-	-	-	-	RDCTMOTP_EN	00	
95	CTM_OTP	R/W	-	-	-	OTP_WR_FLAG	OTP_RD_LENGTH[3:0]				0E
96		R/W	CTM_OTP0[7:0]							00	
97		R/W	CTM_OTP1[7:0]							00	
98		R/W	CTM_OTP2[7:0]							00	
99		R/W	CTM_OTP3[7:0]							00	
9A		R/W	CTM_OTP4[7:0]							00	
9B		R/W	CTM_OTP5[7:0]							00	
9C		R/W	CTM_OTP6[7:0]							00	
9D		R/W	CTM_OTP7[7:0]							00	
9E		R/W	CTM_OTP8[7:0]							00	
9F		R/W	CTM_OTP9[7:0]							00	
A0		R/W	CTM_OTP10[7:0]							00	
A1		R/W	CTM_OTP11[7:0]							00	
A2		R/W	CTM_OTP12[7:0]							00	
A3		R/W	CTM_OTP13[7:0]							00	
A4		R/W	CTM_OTP14[7:0]							00	
A5		R/W	CTM_OTP15[7:0]							00	

JD9365D

A6	SETCUSTM_ID	R/W	-	-	-	-	-	-	-	00	
A7		R/W	CUSTM1[7:0]								FF
A8		R/W	CUSTM2[7:0]								FF
A9		R/W	CUSTM3[7:0]								FF
AA		R/W	CUSTM4[7:0]								FF
AB		R/W	CUSTM5[7:0]								FF
AC		R/W	CUSTM6[7:0]								FF
AD		R/W	CUSTM7[7:0]								FF
AE		R/W	CUSTM8[7:0]								FF
AF	SET_TE_PA	R/W	-	-	-	VHS_TEO_N_OPT	TP_VS_EN2	TP_VS_WIDTH2[10:8]			0D
B0	D2	R/W	TP_VS_WIDTH2[7:0]								00
B1		R/W	TP_VS_SHIFT2[7:0]								01

TABLE 2.3: JADARD PAGE 1 COMMAND LIST

JD9365D

2.4. All page command description

241 SET_PAGE(ALL Page-RE0h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RE0h	R/W	-	-	-	-	-	PAGE[2:0]			
Description	This command is used to change page. PAGE[2:0]: Register page select.									
Restriction	-									

JD9365D

242 SET_PASSWORD(ALL Page-RE1h~RE3h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
Command	R/W	1	0	1	1	1	1	1	1	BF
RE1h	R/W	PASSWD1[7:0]								
RE2h	R/W	PASSWD2[7:0]								
RE3h	R/W	PASSWD3[7:0]								
Description	This command is used to set password to access inhouse register. When {PASSED1, PASSWD2, PASSWD3}= {93h, 65h, F8h} => Enable STD CMD and all in-house registers {5Ah, A5h, FFh} => Enable STD CMD and Page0~3 registers {09h, B1h, 7Fh} => No access permission {69h, 96h, 55h} =>release no access permission, only STD CMD Others => Enable STD CMD only (Default)									
Restriction	-									

JD9365D

243. AUTO_DISP_SETTING(ALL Page -RE4h~RE5h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX																				
RE4h	R/W	-	-	-	-	GON	DTE	DISP[1:0]																						
RE5h	R/W	-	-	DISP_SW_OPT[1:0]		-	BLK_DISABLE	BLK_MODE[1:0]																						
Description	This command is used to set display sequence Related Setting.																													
	DISP[1:0]: Source output setting.																													
	<table><tr><td>DISP1</td><td>DISP0</td><td>Internal Display Operations</td><td>Source Output</td></tr><tr><td>0</td><td>0</td><td>Halt</td><td>VSSD</td></tr><tr><td>0</td><td>1</td><td>Inhibit</td><td>Inhibit</td></tr><tr><td>1</td><td>0</td><td>Operate</td><td>V255</td></tr><tr><td>1</td><td>1</td><td>Operate</td><td>Display</td></tr></table>										DISP1	DISP0	Internal Display Operations	Source Output	0	0	Halt	VSSD	0	1	Inhibit	Inhibit	1	0	Operate	V255	1	1	Operate	Display
	DISP1	DISP0	Internal Display Operations	Source Output																										
	0	0	Halt	VSSD																										
	0	1	Inhibit	Inhibit																										
	1	0	Operate	V255																										
	1	1	Operate	Display																										
	DTE, GON: Gate / SD output setting.																													
	<table><tr><td>GON</td><td>Gate Output</td></tr><tr><td>0</td><td>VGL</td></tr><tr><td>1</td><td>VGH/VGL</td></tr></table>										GON	Gate Output	0	VGL	1	VGH/VGL														
GON	Gate Output																													
0	VGL																													
1	VGH/VGL																													
<table><tr><td>DTE</td><td>SD Output</td></tr><tr><td>0</td><td>Off (Hi-Z)</td></tr><tr><td>1</td><td>ON</td></tr></table>										DTE	SD Output	0	Off (Hi-Z)	1	ON															
DTE	SD Output																													
0	Off (Hi-Z)																													
1	ON																													
BLK_DISABLE: When this bit is 1, no blanking frame is displayed after SLPIN/SLPOUT. When this bit is 0, 2blanking frame are displayed after SLPIN/SLPOUT.																														
BLK_MODE[1:0]: When BLK_DISABLE=0,																														
<table><tr><td>BLK_MODE [1:0]</td><td>Blank mode</td></tr><tr><td>00</td><td>Turn off blanking after 2 blanking frame</td></tr><tr><td>01</td><td>Keep blanking after 2 blanking frames</td></tr><tr><td>10</td><td>Inhibit</td></tr><tr><td>11</td><td>Inhibit</td></tr></table>										BLK_MODE [1:0]	Blank mode	00	Turn off blanking after 2 blanking frame	01	Keep blanking after 2 blanking frames	10	Inhibit	11	Inhibit											
BLK_MODE [1:0]	Blank mode																													
00	Turn off blanking after 2 blanking frame																													
01	Keep blanking after 2 blanking frames																													
10	Inhibit																													
11	Inhibit																													
DISP_SW_OPT[1:0]: Define status of DISP[1:0] after SW reset command received.																														
Restriction	-																													

JD9365D

244. SET_WD: Setup watch dog (ALL Page -RE6h~RE8h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX																														
RE6h	R/W	-				-	-	WD_MODE[1:0]																																
RE7h	R/W	-	-	-	WD_off	WD_Timer[3:0]																																		
RE8h	R/W	-	-	-	-	-	-	WD_FMS	WD_FBLK																															
Description	WD_MODE[1:0]: Watch dog mode select.																																							
	<table><tr><td colspan="2">WD_MODE[1:0]</td><td colspan="2">WD Alarm start</td><td colspan="2">WD Alarm stop</td></tr><tr><td>0</td><td>0</td><td colspan="2">Inhibit</td><td colspan="2">Inhibit</td></tr><tr><td>0</td><td>1</td><td colspan="2">Inhibit</td><td colspan="2">Inhibit</td></tr><tr><td>1</td><td>0</td><td colspan="2">Display off</td><td colspan="2">Display on</td></tr><tr><td>1</td><td>1</td><td colspan="2">Inhibit</td><td colspan="2">Inhibit</td></tr></table>										WD_MODE[1:0]		WD Alarm start		WD Alarm stop		0	0	Inhibit		Inhibit		0	1	Inhibit		Inhibit		1	0	Display off		Display on		1	1	Inhibit		Inhibit	
	WD_MODE[1:0]		WD Alarm start		WD Alarm stop																																			
	0	0	Inhibit		Inhibit																																			
	0	1	Inhibit		Inhibit																																			
	1	0	Display off		Display on																																			
	1	1	Inhibit		Inhibit																																			
	WD_off: Bit=1, disable watch dog function.																																							
	WD_Timer[3:0]: Set watch dog timer. When external HSYNC stop, watch dog timer start to count. Send alarm signal to watch dog if timer counting value is the same as WD_Timer[3:0] setting. When HSYNC resume, alarm signal is stoped.																																							
	<table><tr><td colspan="2">WD_Timer[3:0]</td><td colspan="2">Time to alarm</td></tr><tr><td colspan="2">0x00</td><td colspan="2">0 x 512osc(Inhibit)</td></tr><tr><td colspan="2">0x01</td><td colspan="2">1 x 512osc</td></tr><tr><td colspan="2">:</td><td colspan="2">:</td></tr><tr><td colspan="2">0x0E</td><td colspan="2">14 x 512osc</td></tr><tr><td colspan="2">0x0F</td><td colspan="2">15 x 512osc</td></tr></table>										WD_Timer[3:0]		Time to alarm		0x00		0 x 512osc(Inhibit)		0x01		1 x 512osc		:		:		0x0E		14 x 512osc		0x0F		15 x 512osc							
WD_Timer[3:0]		Time to alarm																																						
0x00		0 x 512osc(Inhibit)																																						
0x01		1 x 512osc																																						
:		:																																						
0x0E		14 x 512osc																																						
0x0F		15 x 512osc																																						
WD_FMS: Force Mode Switch when WDT occurs(not release)																																								
WD_FBLK: Force Blanking when WDT or Hwreset occurs(not release)																																								
Restriction	-																																							

JD9365D

245. OTP PROG: OTP Program (ALL Page -RE9h~REFh)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RE9h	R/W	OTP_MASK[7:0]								
REAh	R/W	X	X	X	X	X	OTP_INDEX[10:8]			
REBh	R/W	OTP_INDEX[7:0]								
RECh	R/W	X	X	X	OTP_READ_SINGLE	OTP_WRITE_SINGLE	OTP_LOAD_DISABLE	OTP_INT_VPP	OTP_AUTO_PROG	
REDh	R	OTP_RDATA[7:0]								
REEh	R/W	OTP_WDATA[7:0]								
REFh	R/W	OTP_PROG	OTP_VPP_SEL	OTP_PCE	OTP_PWE	X	X	OTP_PTM[1:0]		
Description	OTP_MASK[7:0]: Bit programming mask, if 1, means this bit can't be programmed. OTP_INDEX[10:0] : Set index of OTP table for programming. OTP_WRITE_READ_SINGLE: Read 1 byte OTP value. 0: Disable. 1: Enable. EX: set INDEX[9:0], READ_SINGLE=1 and AUTO_PROG=1, then read OTP_RDATA[7:0] OTP_WRITE_PROG_SINGLE: For 1 byte OTP programing only.(Not release to customer) 0: Disable. 1: Enable. EX: set MASK[7:0], INDEX[9:0] ,WDATA[7:0] and WRITE_SINGLE=1 first then set AUTO_PROG=1, the WDATA will write to INDEX. OTP_LOAD_DISABLE: Auto reload OTP value. 0: Enable auto reload OTP value after SLPOUT command send. 1: Disable auto reload OTP value after SLPOUT command send. OTP_INT_VPP: Select VPP voltage from external or internal. 0: VPP voltage from external power supply. 1: VPP voltage from internal power supply (VGH). OTP_AUTO_PROG: When set to 1, the register content of OTP index is programmed. OTP_RDATA [7:0]: Read back the OTP index data. (Not release to customer) Following description is for OTP manual mode. (Not release to customer) OTP_WDATA: Set OTP data for programming when OTP_PROG_SINGLE=1.									

JD9365D

	OTP_PPORG: Program mode enable. 0:disable, 1:enable. OTP_VPP_SEL: When written to 1, VPP voltage is fed to OTP. OTP_POR: Pin for read operation OTP_PWE: Define program cycle. 1: means OTP is able to be programmed. OTP_PTM[1:0]: Test mode.
Restriction	-

JD9365D

24.6. SELF_DIAG(ALL Page -RF0h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RF0h	R	regload_detec	func_detc	attach_detec	gbreak_detec	-	-	-	-	HEX
Description	This command is used to for up debug									
Restriction	-									

JD9365D

2.5. Jadard page 0 command description

25.1 RDCUSTMID: Read Customer ID (Page0-R1Bh)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R1Bh	R	CUSTM1[7:0]								
	R	CUSTM2[7:0]								
	R	CUSTM3[7:0]								
	R	CUSTM4[7:0]								
	R	CUSTM5[7:0]								
	R	CUSTM6[7:0]								
	R	CUSTM7[7:0]								
	R	CUSTM7[7:0]								
Description	This command is used to read customer ID. Read the value of page1_RA7h to RAEh CUSTM1[7:0] mapping to Page1_RA7h CUSTM2[7:0] mapping to Page1_RA8h CUSTM3[7:0] mapping to Page1_RA9h CUSTM4[7:0] mapping to Page1_RAAh CUSTM5[7:0] mapping to Page1_RABh CUSTM6[7:0] mapping to Page1_RACH CUSTM7[7:0] mapping to Page1_RADh CUSTM8[7:0] mapping to Page1_RAEh									
Restriction	-									

JD9365D

252 SEQUENCE CTRL: Power on sequence control (Page0-R70h~R76h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX																													
R70h	R/W	X	DC0H[2:0]			X	DC1H[2:0]																																
R71h	R/W	X	DC2H[2:0]			X	DC3H[2:0]																																
R72h	R/W	X	X	X	X	X	DC7H[2:0]																																
R73h	R/W	DC0L	DC1L	DC2L	DC3L	DC4L	DC5L	DC6L	DC7L																														
R74h	R/W	T0P[1:0]		T1P[1:0]		T8P[1:0]		X	X																														
R75h	R/W	X	SOFT0H[2:0]			X	SOFT0L[2:0]																																
R76h	R/W	X	X	X	X	X	X	X	POW_ON_STATE																														
Description	This command is used to set power on sequence Related Setting.																																						
	DC0H[2:0]: Select the AVDD power on timing(P1~P8).																																						
	DC1H[2:0]: Select the VCL power on timing(P1~P8).																																						
	DC2H[2:0]: Select the AVEE power on timing(P1~P8).																																						
	DC3H[2:0]: Select the VGL power on timing(P1~P8).																																						
	DC4H[2:0]: Fix the VGH power on timing at P5.																																						
	DC5H[2:0]: Fix the Gamma reference voltage power on timing at P6.																																						
	DC6H[2:0]: Fix the VCOM power on timing at P7.																																						
	DC7H[2:0]: Option.																																						
	<table><tr><th colspan="3">Power-on Sequence option</th></tr><tr><td>DCxH[2:0]</td><td>Go High</td><td></td></tr><tr><td>0</td><td>P1</td><td>DC0H default</td></tr><tr><td>1</td><td>P2</td><td>DC1H default</td></tr><tr><td>2</td><td>P3</td><td>DC2H default</td></tr><tr><td>3</td><td>P4</td><td>DC3H default</td></tr><tr><td>4</td><td>P5</td><td>DC4H fixed</td></tr><tr><td>5</td><td>P6</td><td>DC5H fixed</td></tr><tr><td>6</td><td>P7</td><td>DC6H fixed / DC7H default</td></tr><tr><td>7</td><td>P8</td><td></td></tr></table>										Power-on Sequence option			DCxH[2:0]	Go High		0	P1	DC0H default	1	P2	DC1H default	2	P3	DC2H default	3	P4	DC3H default	4	P5	DC4H fixed	5	P6	DC5H fixed	6	P7	DC6H fixed / DC7H default	7	P8
Power-on Sequence option																																							
DCxH[2:0]	Go High																																						
0	P1	DC0H default																																					
1	P2	DC1H default																																					
2	P3	DC2H default																																					
3	P4	DC3H default																																					
4	P5	DC4H fixed																																					
5	P6	DC5H fixed																																					
6	P7	DC6H fixed / DC7H default																																					
7	P8																																						
DC0L: Select the AVDD power off timing(P10~P11).																																							
DC1L: Select the VCL power off timing(P10~P11)																																							
DC2L: Select the AVEE/VDDN power off timing(P10~P11)																																							
DC3L: Select the VGL power off timing(P10~P11)																																							
DC4L: Select the VGH power off timing(P10~P11)																																							
DC5L: Select the Gamma reference voltage power off timing(P10~P11)																																							
DC6L: Select the VCOM power off timing(P10~P11)																																							

DC7L: Option

Power-off Sequence option		
DCxL	Go Low	
0	P10	DC1-3L,DC5-7L default
1	P11	DC0/4L default

TP0[1:0]: Slecet the period time for T0.

TP1[1:0]: Slecet the period time for T1.

TP8[1:0]: Slecet the period time for T8.

Power-on Period option		
TxP[1:0]	Period	
0	0.5ms	
1	3ms	T0 / T8 default
2	6ms	T1 default
3	10ms	

SOFT0H[2:0]: Select the AVDD soft pump start timing(P0~P7).

Soft Start Option		
SOFTxH[2:0]	Go High	
0	P0	SOFT0H default
1	P1	
2	P2	
3	P3	
4	P4	
5	P5	
6	P6	
7	P7	

SOFT0L[2:0]: Select the AVDD soft pump stop timing(P1~P8).

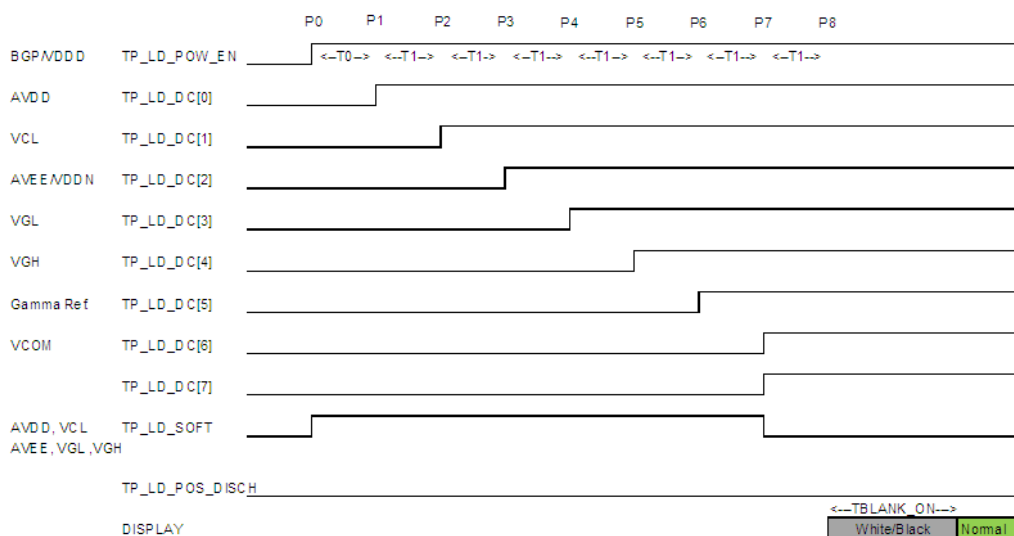
Soft End Option		
SOFTxL[2:0]	Go Low	
0	P1	
1	P2	
2	P3	
3	P4	
4	P5	

JD9365D

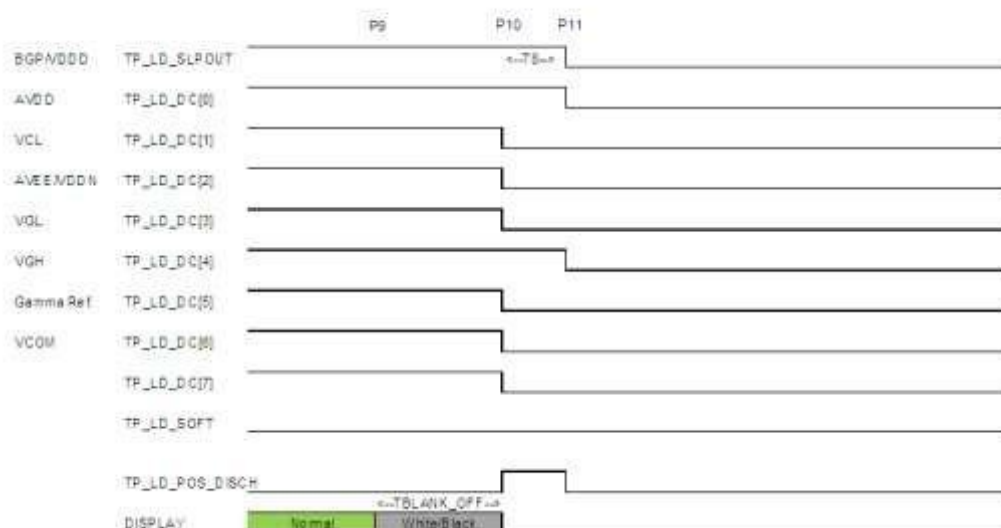
5	P6	
6	P7	SOFT0L default
7	P8	

POW_ON_STATE: All power ready when this bit is high.

• Power-on Sequence Flow



• Power-off Sequence Flow



Restriction -

JD9365D

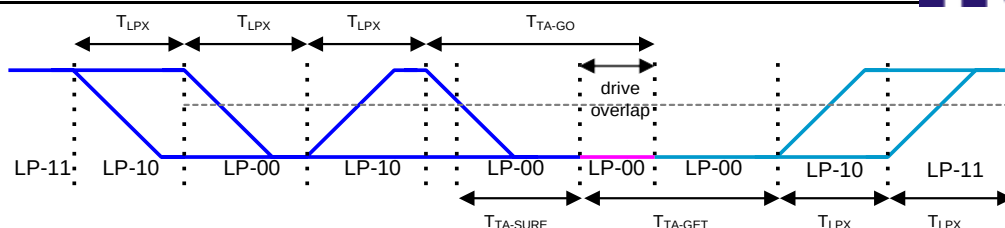
253. SET ID: SETID(OTP*5) (Page0-R77h~R7Ch)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R77h	R	X	X	X	X	X	X	X	X	
R78h	R/W	ID1[7:0]								
R79h	R/W	ID2[7:0]								
R7Ah	R/W	ID3[7:0]								
R7Bh	R/W	ID4[7:0]								
R7Ch	R	X	X	X	X	X	ID_OTP_TIMES[2:0]			
Description	ID1[7:0]: This is used to set ID1 and read value from DAh. ID2[7:0]: This is used to set ID2 and read value from DBh. ID3[7:0]: This is used to set ID3 and read value from DCh. ID4[7:0]: This is used to set ID4. ID_OTP_TIMES [2:0]: Read the OTP program times of ID. (Not release to customer) 0: No valid SETID in OTP, use register default value. 1: SETID is programmed once in OTP. Reload 1st valid one. 5: SETID is programmed 5 times in OTP. Reload 5th valid one.									
Restriction	-									

JD9365D

254. SETDSISETUP: Set DSI Setup (Page0-R80h~R82h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R80h	R/W	X	DSI_INIT0[6:0]							
R81h	R/W	DSI_INIT1[7:0]								
R82h	R/W	DSI_INIT2 [7:0]								
Description	This command is used to set DSI Related Setting.									
	//DSI_INIT0									
	//DSI_INIT0[1:0]: Lane Num.									
	2'b00		1 lane							
	2'b01		2 lane							
	2'b10		3 lane							
	2'b11		4 lane							
	DSI_INIT0[3:2]: VC_ID, virtual channel identifier of RX.Default is 2'b00.									
	DSI_INIT0[4]: EoT_pkt_en.									
	0: Rx doesn't care the EOT packet (default)(protocol_error disable), 1: Rx must receive the EOT packet before entering the stop state									
DSI_INIT0[5]: ERR_rpt_en.										
0: disable the all error detector 1: enable (default)										
//DSI_INIT1										
DSI_INIT1[1:0]: TX_s_wait. T _{TA_SURE} + T _{TA_GET} (spec. definition:min is 6 T _{LPX} , max is 7 T _{LPX} .)										
Note:T _{TA_SURE} time is fixed by osc_on_time+OSC_div_setup_time , the TX_s_wait only affect the T _{TA_GET} time										
2'b00		7T _{LPX}								
2'b01		8T _{LPX}								
2'b10		9T _{LPX}								
2'b11		12T _{LPX}								
DSI_INIT1[4:2]: TX_e_wait. T _{TA-GO} (spec. definition: 4 T _{LPX} .)										
3'b000		1T _{LPX}								
3'b001		2T _{LPX}								
3'b010		3T _{LPX}								
3'b011		4T _{LPX}								
3'b100		5T _{LPX}								
3'b101		6T _{LPX}								
3'b110		7T _{LPX}								
3'b111		8T _{LPX}								



DSI_INIT1 [6:5]: TX_OSC_DIV[1:0].

2'b00	OSCM_CLK_DIV2
2'b01	OSCM_CLK_DIV3
2'b10	OSCM_CLK_DIV4
2'b11	OSCM_CLK

DSI_INIT1[7]: CD_disable.

0: Enable contention detector flag(default).

1: disable

//DSI_INIT2

DSI_INIT2[0]: RST_trigger_EN.

0: Disable the reset function of the escape entry code.

1: Enable(Whole chip level), function as SW_RST. (default)

DSI_INIT2[1]: ECC_dis.

1: Ignore the ECC error report function, but the ECC check function still works (default),

0: normal function

DSI_INIT2[2]: CRC_chk_en.

0: Don't report the CRC error.

1: Report the CRC error. (default)

Restriction

-

JD9365D

255. SETDSISETUP: Set DSI Setup2 (Page0-R83h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX	
R83h	R/W	X	DSI_INIT3[6:0]								
Description	This command is used to set DSI Related Setting. //DSI_INIT3 DSI_INIT3[2:0]: DSWAP[2:0]. Below tabl show the mapping while PN is not swap.										
	DSWAP [2:0]	HS_ D0N	HS_ D0P	HS_ D1N	HS_ D1P	HS_ CN	HS_ CP	HS_ D2N	HS_ D2P	HS_ D3N	HS_ D3P
	3'b000	D2-	D2+	D1-	D1+	CLK-	CLK+	D0-	D0+	D3-	D3+
	3'b001	D2-	D2+	D0-	D0+	CLK-	CLK+	D1-	D1+	D3-	D3+
	3'b010	D3-	D3+	D1-	D1+	CLK-	CLK+	D0-	D0+	D2-	D2+
	3'b011	D3-	D3+	D0-	D0+	CLK-	CLK+	D1-	D1+	D2-	D2+
	3'b100	D3-	D3+	D2-	D2+	CLK-	CLK+	D1-	D1+	D0-	D0+
	3'b101	D3-	D3+	D1-	D1+	CLK-	CLK+	D2-	D2+	D0-	D0+
	3'b110	D0-	D0+	D2-	D2+	CLK-	CLK+	D1-	D1+	D3-	D3+
	3'b111	D0-	D0+	D1-	D1+	CLK-	CLK+	D2-	D2+	D3-	D3+
	DSI_INIT3 [3]: PN_SWAPX. 1: PN no swap(default). 0: swap.										
Restriction	-										

JD9365D

25.6. UP_START_CTRL (Page0-R84h~R85h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R84h	R/W	UP_KEY1[7:0]								
R85h	R/W	UP_KEY2[7:0]								
Description	OTP_KEY1/KEY2[7:0]: This command is used to enter or leave OTP program mode. Before OTP programming, setting OTP_KEY1=5Ah and OTP_KEY2=A5h first. Otherwise OTP programming will fail.									
	OTP_KEY1[7:0]		OTP_KEY2[7:0]		Description					
	0x5Ah		0xA5h		Enter OTP program mode					
	0x00h		0x00h		Leave OTP program mode					
Restriction	-									

JD9365D

25.7. ALS: SET_ALS (Page0-R88h~R8Bh)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R88h	R/W	ALS[31:24]								
R89h	R/W	ALS[23:16]								
R8Ah	R/W	ALS[15:08]								
R8Bh	R/W	ALS[07:00]								
Description	32bits ambient value (Unit: LUX) Use for “SLR Auto” mode. IEC 0x7									
Restriction	-									

JD9365D

2.6. Jadard page 1 command description

261. VCOM_SET: Set VCOM Voltage (Page1-R00h~R02h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX																										
R00h	R/W	X							VCOM[8]																											
R01h	R/W	VCOM[7:0]																																		
R02h	R/W	X					VCOM_OTP_TIMES[2:0]																													
Description	This command is used to set VCOM voltage and read OTP writing times for VCOM. VCOM[8:0]: Set VCOM voltage when GIP normal scan. <table border="1"> <thead> <tr> <th>VCOM[8:0]</th> <th>VCOM (V)</th> </tr> </thead> <tbody> <tr><td>000h</td><td>-0.30</td></tr> <tr><td>001h</td><td>-0.31</td></tr> <tr><td>002h</td><td>-0.32</td></tr> <tr><td>003h</td><td>-0.33</td></tr> <tr><td>004h</td><td>-0.34</td></tr> <tr><td>:</td><td>-10mv/step</td></tr> <tr><td>171h</td><td>-3.99</td></tr> <tr><td>172h</td><td>-4.00</td></tr> <tr><td>173h~1FCh</td><td>Inhibit</td></tr> <tr><td>1FDh</td><td>VCOMR</td></tr> <tr><td>1FEh</td><td>AVSS</td></tr> <tr><td>1FFh</td><td>Hi-Z</td></tr> </tbody> </table> VCOM_OTP_TIMES[2:0]: This register just can be written by up. Due to VCOM is belonging to MTP, VCOM_OTP_TIMES[2:0] is used to record OTP being programmed times of VCOM.										VCOM[8:0]	VCOM (V)	000h	-0.30	001h	-0.31	002h	-0.32	003h	-0.33	004h	-0.34	:	-10mv/step	171h	-3.99	172h	-4.00	173h~1FCh	Inhibit	1FDh	VCOMR	1FEh	AVSS	1FFh	Hi-Z
	VCOM[8:0]	VCOM (V)																																		
	000h	-0.30																																		
	001h	-0.31																																		
	002h	-0.32																																		
	003h	-0.33																																		
	004h	-0.34																																		
	:	-10mv/step																																		
	171h	-3.99																																		
	172h	-4.00																																		
	173h~1FCh	Inhibit																																		
	1FDh	VCOMR																																		
	1FEh	AVSS																																		
	1FFh	Hi-Z																																		
	Restriction	-																																		

JD9365D

262 VCOM_R_SET: Set VCOM Voltage when reverse scan (Page1-R03h~R05h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R03h	R/W	X							VCOM_R[8]	
R04h	R/W	VCOM_R[7:0]								
R05h	R	X					VCOM_R_OTP_TIMES[2:0]			
Description	This command is used to set VCOM voltage and read OTP writing times for VCOM.									
	VCOM_R[8:0]: Set VCOM voltage when GIP reverse scan.									
	VCOM_R[8:0]					VCOM (V)				
	000h					-0.30				
	001h					-0.31				
	002h					-0.32				
	003h					-0.33				
	004h					-0.34				
	:					-10mv/step				
	171h					-3.99				
	172h					-4.00				
	173h~1FCh					Inhibit				
	1FDh					VCOMR				
	1FEh					AVSS				
1FFh					Hi-Z					
VCOM_R_OTP_TIMES[2:0]: This register just can be written by up.										
Due to VCOM is belonging to MTP, VCOM_R_OTP_TIMES[2:0] is used to record OTP being programmed times of VCOM_R.										
Restriction	-									

JD9365D

263. PWRIC SET: External Power IC timing setting (Page1-R0Ah~R0Eh)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX																																																												
R0Ah	R/W	X	X	JD_RELD	VCSW_SWAP	ENVREF_S[3:0]																																																																
R0Bh	R/W	PWRIC_SET[7:0]																																																																				
R0Ch	R/W	X	PWRIC_CLK1[2:0]			X	PWRIC_IDLE_CLK1[2:0]																																																															
R0Dh	R/W	HX_SOFT	PWRIC_CLK2[2:0]			PWRIC_HXCLK[3:0]																																																																
R0Eh	R/W	X	X	X	X	X	X	LED_PWR_S[1:0]																																																														
Description	This command is used to set frequency of charge/pump of AVDD/AVEE for external Power IC. JD_RELD: During blanking area, driver ic will re-send new pumping ratio to external power ic(JD5xxx), set this bit 1 to enable this function. VCSW_SWAP: Swap VCSW1 and VCSW2 for power ic, set this bit 1 to enable this function. ENVREF_S[3:0]: Set voltage level of VCSW1 to JD power IC. The register is valid at DCDCM[2:0]=001. <table> <tr> <th>ENVREF_S[3:0]</th> <th>ENVREF(V)</th> <th>ENVREF_S[3:0]</th> <th>ENVREF(V)</th> </tr> <tr><td>0h</td><td>1.55</td><td>7h</td><td>1.90</td></tr> <tr><td>1h</td><td>1.60</td><td>8h</td><td>1.95</td></tr> <tr><td>2h</td><td>1.65</td><td>9h</td><td>2.00</td></tr> <tr><td>3h</td><td>1.70</td><td>Ah</td><td>2.05</td></tr> <tr><td>4h</td><td>1.75</td><td>Bh</td><td>2.10</td></tr> <tr><td>5h</td><td>1.80</td><td>Ch</td><td>2.15</td></tr> <tr><td>6h</td><td>1.85</td><td>Other</td><td>Inhibitt</td></tr> </table> PWRIC_SET[7:0]: Set reference clock number to detect pumping ratio of JD power IC. The register is valid at DCDCM[2:0]=001. <table> <tr> <th>PWRIC_SET[7:0]</th> <th>AVDD / AVEEPumping ratio</th> </tr> <tr><td>0 reference clocks</td><td>2 x</td></tr> <tr><td>9 reference clocks</td><td>3 x</td></tr> <tr><td>17 reference clocks</td><td>1.5 x</td></tr> </table> PWRIC_CLK1 [2:0]: Set frequency of pumping clock to JD pwoer IC at normal mode operation. PWRIC_IDLE_CLK1 [2:0]: Set frequency of pumping clock to JD pwoer IC at Idle mode operation. The registers are valid at DCDCM[2:0]=001. <table> <tr> <th>PWRIC_CLK1[2:0]/ PWRIC_IDLE_CLK1[2:0]</th> <th>VCSW2</th> <th>PWRIC_CLK1[2:0]/ PWRIC_IDLE_CLK1[2:0]</th> <th>VCSW2</th> </tr> <tr><td>0h</td><td>HSCLK / 16</td><td>4h</td><td>HSCLK</td></tr> <tr><td>1h</td><td>HSCLK / 8</td><td>5h</td><td>HSCLK x 2</td></tr> <tr><td>2h</td><td>HSCLK / 4</td><td>6h</td><td>HSCLK x 4</td></tr> <tr><td>3h</td><td>HSCLK / 2</td><td>7h</td><td>HSCLK x 8</td></tr> </table> PWRIC_CLK2 [2:0]: internal used, not open. HX_SOFT: internal used, not open.										ENVREF_S[3:0]	ENVREF(V)	ENVREF_S[3:0]	ENVREF(V)	0h	1.55	7h	1.90	1h	1.60	8h	1.95	2h	1.65	9h	2.00	3h	1.70	Ah	2.05	4h	1.75	Bh	2.10	5h	1.80	Ch	2.15	6h	1.85	Other	Inhibitt	PWRIC_SET[7:0]	AVDD / AVEEPumping ratio	0 reference clocks	2 x	9 reference clocks	3 x	17 reference clocks	1.5 x	PWRIC_CLK1[2:0]/ PWRIC_IDLE_CLK1[2:0]	VCSW2	PWRIC_CLK1[2:0]/ PWRIC_IDLE_CLK1[2:0]	VCSW2	0h	HSCLK / 16	4h	HSCLK	1h	HSCLK / 8	5h	HSCLK x 2	2h	HSCLK / 4	6h	HSCLK x 4	3h	HSCLK / 2	7h	HSCLK x 8
	ENVREF_S[3:0]	ENVREF(V)	ENVREF_S[3:0]	ENVREF(V)																																																																		
	0h	1.55	7h	1.90																																																																		
	1h	1.60	8h	1.95																																																																		
	2h	1.65	9h	2.00																																																																		
	3h	1.70	Ah	2.05																																																																		
	4h	1.75	Bh	2.10																																																																		
	5h	1.80	Ch	2.15																																																																		
	6h	1.85	Other	Inhibitt																																																																		
	PWRIC_SET[7:0]	AVDD / AVEEPumping ratio																																																																				
0 reference clocks	2 x																																																																					
9 reference clocks	3 x																																																																					
17 reference clocks	1.5 x																																																																					
PWRIC_CLK1[2:0]/ PWRIC_IDLE_CLK1[2:0]	VCSW2	PWRIC_CLK1[2:0]/ PWRIC_IDLE_CLK1[2:0]	VCSW2																																																																			
0h	HSCLK / 16	4h	HSCLK																																																																			
1h	HSCLK / 8	5h	HSCLK x 2																																																																			
2h	HSCLK / 4	6h	HSCLK x 4																																																																			
3h	HSCLK / 2	7h	HSCLK x 8																																																																			

JD9365D

	PWRIC_HXCLK[3:0]: internal used, not open. LED_PWR_S[1:0]: internal used, not open
Restriction	-

fitipower confidential

JD9365D

264. GAMMA_SET: Set Gama Reference Voltage(Page1-R17h~R1Ch)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R17h	R/W	X			VGMP_S[8]	X			VGSP_S[8]	
R18h	R/W	VGMP_S[7:0]								
R19h	R/W	VGSP_S[7:0]								
R1Ah	R/W				VGMN_S[8]	X			VGSN_S[8]	
R1Bh	R/W	VGMN_S[7:0]								
R1Ch	R/W	VGSN_S[7:0]								
Description	This command is used to set postive / Negative reference volatge of Gamma.									
	VGMP_S[8:0]: Set power source volatge of postive polarity.									
	VGMP_S [8:0]					VGMP (V)				
	0h					0 (GND)				
	:					Inhibit				
	27h					2.6				
	28h					2.6125				
	:					12.5mv/step				
	11Eh					5.6875				
	11Fh					5.7				
Others					Inhibit					
VGSP_S [8:0]: Set reference volatge of postive polarity.										
VGMP_S [8:0]					VGSP (V)					
0h					0 (GND)					
1h					0.3					
2h					0.3125					
:					12.5mv/step					
110h					3.6875					
111h					3.7					
Others					Inhibit					
VGMN_S [8:0]: Set power source volatge of negative polarity.										
VGMP_S [8:0]					VGMN (V)					
0h					0 (GND)					
:					Inhibit					
27h					-2.6					
28h					-2.6125					
:					12.5mv/step					
11Eh					-5.6875					
11Fh					-5.7					
Others					Inhibit					
VGSN_S [8:0]: Set reference volatge of negative polarity.										
VGMP_S[8:0]					VGSN (V)					
0h					0 (GND)					
1h					-0.3					
2h					-0.3125					
:					12.5mv/step					
110h					-3.6875					
111h					-3.7					



JD9365D

		Others	Inhibit	
Restriction	-			

fitipower confidential

JD9365D

265. OTP_SET(Page1-R1Dh~R1Eh)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R1Dh	R/W	-	-	VPP_DT[1:0]		-	VPP_VGH_RT[2:0]		-	
R1Eh	R/W	-	VPP_VGH_S[6:0]						-	
Description	This command is used to set internal OTP program related setting.									
	VPP_DT[1:0]: When INT_VPP=1(ECh), internal OTP will program 2bit step by step until 1 byte program finished. VPP_DT[1:0] can define delay time between each cycle.									
	VPP_DT[2:0]		Delaytime (ms)							
	0h		0.25							
	1h		0.5							
	2h		1							
	3h		2							
	VPP_VGH_RT[2:0]: Set VGH DC/DC charge/pump ratio of internal OTP program.									
	VPP_VGH_RT[2:0]		Charge pump Ratio							
	0		AVDD-VCL							
1		2*AVDD								
2		2*AVDD-VCL								
3		2*AVDD-AVEE								
4		3*AVDD-VCL								
5		3*AVDD-AVEE								
6		3*AVDD-AVEE-VCL								
7		3*AVDD-2*AVEE								
VPP_VGH_S[6:0]: Set VGH voltage level of internal OTP program.										
VPP_VGH_S[6:0]		VGH (V)								
000_0011		7.6								
Others		Inhibit								
Restriction	-									

JD9365D

266. GATE_POWER_SET: Set GIP Output High Level (Page1-R1Fh~R22h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R1Fh	R/W	X	VGH_REG[6:0]							
R20h	R/W	X	VGL_REG[5:0]							
R21h	R/W	X	VGL_REG2[5:0]							
R22h	R/W		VGH_REG_SHT	VGL_REG_SHT	VGL_REG2_SHT	VGH_REG_EN	VGL_REG_EN	VGL_REG2_EN	SHT_VGL_REG	

Description

This command is used to set GIP output voltage level.

VGH_REG[6:0]: Set VGH regulator volatge level.

VGH_REG[6]	VGH_REG[5:0]	VGH_REG(V)
0	00h	2.6
0	01h	2.8
0	02h	3.0
:	:	200mv/step
0	3Dh	14.8
0	3Eh	15.0
0	3Fh	15.2
1	00h	6.6
1	01h	6.8
1	02h	7.0
:	:	200mv/step
1	3Dh	18.8
1	3Eh	19.0
1	3Fh	19.2

VGL_REG[5:0]: Set VGL_REG regulator volatge level.

VGL_REG2[5:0]: Set VGL_REG2 regulator volatge level.

VGL_REG[5:0] / VGL_REG2[5:0]	VGL_REG(V) / VGL_REG2(V)
00h	-3.0
01h	-3.2
02h	-3.4
:	200mv/step
3Dh	-15.2
3Eh	-15.4
3Fh	-15.6

VGH_REG_SHT: Define V(VGH_REG) conenct to VGH.

VGL_REG_SHT: Define V(VGL_REG) conenct to VGL.

VGL_REG2_SHT: Define V(VGL_REG2) conenct to VGL.

XX_SHT	Connect V(VGH_REG),V(VGL_REG)/V(VGL_REG2) to VGH/VGL
0	Disbale
1	Enable

VGH_REG_EN: Set VGH_REG output state.

VGL_REG_EN: Set VGL_REG output state.

VGL_REG2_EN: Set VGL_REG2 output state.

XX_EN	Output Level
1	Regulator
0	Hi-Z

JD9365D

	SHT_VGL_REG: Set connect V(VGL_REG2) to V(VGL_REG). 0: Disbale 1: Enable			
	VGL_REG2_PAD		VGL_REG2_EN	
	SHT_VGL_REG	VGL_REG2_SHT	1	0
	0	0	VGL_REG2	Hi-Z
	0	1	VGL	Hi-Z
	1	X	VGL_REG	VGL_REG
Restriction	-			

JD9365D

26.7. DCDC_CTRL(Page1-R23h~R24h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R23h	R/W	-	-	-	-	VGH_NC	VGL_NC	VCLREG_OP_CEN	VCL_NC	-
R24h	R/W	VGH_EN	VGL_EN	AVDD_EN	AVEE_EN	VCL_EN	VDDA_EN	VCLREG_G_EN	VCL2_VDDN	-
Description	This command is used to control engineer mode of power related setting.									
	VGH_NC: Set VGH output state.									
	VGH_NC		VGH output							
	0		Clamping							
	1		No clamping							
	VGL_NC: Set VGL otuptut state									
	VGL_NC		VGH output							
	0		Clamping							
	1		No clamping							
	VCLREG_OP_CEN: VCL stable cap									
	VCLREG_OP_CEN				VCL					
	0				Need stable Cap					
	1				No need stable Cap					
	VCL_NC: Set VCL otuptut state									
	VCL_NC		VGH output							
	0		Clamping							
	1		No clamping							
	VGH_EN: Control VGH DC/DC circuit On/Off.									
VGH_EN		VGH DC/DC circuit								
0		Disbale								
1		Enable								
VGL_EN: Control VGL DC/DC circuit On/Off.										
VGL_EN		VGL DC/DC circuit								
0		Disbale								
1		Enable								
AVDD_EN: Control AVDD DC/DC circuit On/Off.										
AVDD_EN		AVDD DC/DC circuit								
0		Disbale								
1		Enable								
AVEE_EN: Control AVEE DC/DC circuit On/Off.										
AVEE_EN		AVEE DC/DC circuit								
0		Disbale								
1		Enable								
VCL_EN: Control VCL DC/DC circuit On/Off.										
VCL_EN		VCL DC/DC circuit								
0		Disbale								

JD9365D

	<table> <tr> <td>1</td><td>Enable</td></tr> </table> VDDA_EN: Control VDDN DC/DC circuit On/Off. <table> <tr> <td>VDDA_EN</td><td>VDDA DC/DC circuit</td></tr> <tr> <td>0</td><td>Disbale</td></tr> <tr> <td>1</td><td>Enable</td></tr> </table> VCLREG_EN: When VCLEN=1, set VCL circuit source <table> <tr> <td>VCLREG_EN</td><td>VCL Source</td></tr> <tr> <td>0</td><td>VCL from DCDC</td></tr> <tr> <td>1</td><td>VCL from regulator</td></tr> </table> VCL2VDDN: Control source of VDDN output. <table> <tr> <td>VCL2VDDN</td><td>VDDN</td></tr> <tr> <td>0</td><td>VDDN DC/DC Circuit</td></tr> <tr> <td>1</td><td>VCL</td></tr> </table>	1	Enable	VDDA_EN	VDDA DC/DC circuit	0	Disbale	1	Enable	VCLREG_EN	VCL Source	0	VCL from DCDC	1	VCL from regulator	VCL2VDDN	VDDN	0	VDDN DC/DC Circuit	1	VCL
1	Enable																				
VDDA_EN	VDDA DC/DC circuit																				
0	Disbale																				
1	Enable																				
VCLREG_EN	VCL Source																				
0	VCL from DCDC																				
1	VCL from regulator																				
VCL2VDDN	VDDN																				
0	VDDN DC/DC Circuit																				
1	VCL																				
Restriction	-																				

JD9365D

268. POWAMP_CTRL: Set Power Function Related Setting (Page1-R25h~R27h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R25h	R/W	X	AP[2:0]			X				
R26h	R/W	GAS_EN	VCOM_EN	GAS_VCOM_EN	VGM_EN	GAS_AVDD_EN	GAS_AVEE_EN	EN_VDDD_VCI	EN_VDDD_IOVCC	
R27h	R/W	UP_VDD_1P8_EN	VDDD_S1[2:0]			X	VDDD_S2[2:0]			
Description	This command is used to control engineer mode of power related setting.									
	AP[2:0]: Adjust the ratio of fixed current for the operational amplifier in the power circuit.									
	AP[2:0]		Biase Current							
	0h		25%							
	1h		50%							
	2h		75%							
	3h		100%							
	4h		125%							
	5h		150%							
	6h		175%							
	7h		200%							
	GAS_EN: Set GAS deetction function On/Off									
	GAS_EN		GAS deetction							
	0		Disable							
	1		Enable							
VCOM_EN: Control VCOM regulator On/Off										
VCOM_EN		VCOM								
0		Disable								
1		Enable								
GAS_VCOM_EN: Control VCOM output statue during abormal power off.										
GAS_VCOM_EN		VCOM ouput								
0		AVSS								
1		Keep VCOM output								
VGM_EN: Control Postive/Negative regulator of Source output On/Off.										
VGM_EN		VGMP/VGSP VGMN/VGSN								
0		Disable								
1		Enable								
GAS_AVDD_EN: Enable AVDD abnormal power off detect function.										
GAS_AVEE_EN: Enable AVEE abnormal power off detect function.										
UP_VDDD_1P8_EN: internal used,not open										
EN_VDDD_VCI:Control VDDD_VCI regualtor On/Off										
EN_VDDD_IOVCC: Control VDDD_IOVCC regualtor On/'Off										

JD9365D

	VDDD_S1[2:0]: Set VDDD when EN_VDDD_IOVCC=1 VDDD_S2[2:0]: Set VDDD when EN_VDDD_VCI=1
Restriction	-

fitipower confidential

JD9365D

269. SETSTBA: Set Source Option (Page1- R35h~R36h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R35h	R/W	X	GAP[2:0]			SAP[3:0]				
R36h	R/W	SAP_Temp[3:0]				SAP2[3:0]				
Description	This command is used to panel related setting.									
	GAP[2:0]: Set Gamma OP output driving ability. The output current is multiple of fixed current(IREF).									
	GAP[2:0]		Biase current							
	0h		1*IREF							
	1h		2*IREF							
	2h		3*IREF							
	3h		4*IREF							
	4h		5*IREF							
	5h		6*IREF							
	6h		7*IREF							
	7h		8*IREF							
	SAP[3:0]/SAP2[3:0]: Set channel OP output driving ability. The otuptu current is multiple of fixed current(IREF).									
	SAP[3:0]/SAP2[3:0]		Biase current							
	0h		1*IREF							
	1h		2*IREF							
	2h		3*IREF							
	3h		4*IREF							
	4h		5*IREF							
	5h		6*IREF							
	6h		7*IREF							
	7h		8*IREF							
	8h		9*IREF							
	9h		10*IREF							
	Ah		11*IREF							
	Bh		12*IREF							
Ch		13*IREF								
Dh		14*IREF								
Eh		15*IREF								
Fh		16*IREF								
SAP_Temp[3:0]: internal used, not open										
Restriction	-									

JD9365D

2610. SETPANEL: Set Panel Related (Page1-R37h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R37h	R/W	X	Z_line	ENZ[1:0]		SS_PANEL	GS_PANEL	REV_PANEL	BGR_PANEL	
Description	This command is used to panel related setting.									
	Z_line: Set SZ output line									
	Z_line		SZ otuptu line							
	0		Odd line(1 st line)							
	1		Even line(2 nd line)							
	ENZ[1:0]: Set SZ output of Zag-Zag inversion.									
	ENZ[1:0]		SD Dummy outptu							
	01		SZ[2]							
	10		SZ[3]							
	SS_Panel: Set Source scan output direction.									
	SS_Panel		S/D scan direction							
	0		S1 -> S2400							
	1		S2400 -> S1							
	GS_Panel: Set Gate scan output direction.									
	GS_Panel		G/D scan direction							
	0		Top -> Bottom Scan (G1->G1280)							
	1		Bottom -> Top Scan (G1280 -> G1)							
	Note: Exmaple, resolution is 800RGB x 1280.									
	REV_Panel: Set the display of the same data on both normally-white and normally-black panels.									
	REV_Panel		Panel (Normally-type)							
0		Normal Black panel								
1		Normal White panel								
BGR_Panel: This bit is control the order of <R><G> dot color for module supplier.										
BGR_Panel		Color fliter alignment								
0		<S1><S2><S3> align to <R><G>								
1		<S1><S2><S3> align to <G><R>								
Restriction	-									

JD9365D

2611 SETRGBCYC: Set Display Waveform Cycles of RGB Mode (Page1- R38h~R3Fh)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R38h	R/W	X	interlace_line	X			RGB_JDT[2:0]			
R39h	R/W	RGB_N_EQ1[7:0]								
R3Ah	R/W	RGB_N_EQ2[7:0]								
R3Bh	R/W	RGB_N_EQ2_Temp[7:0]								
R3Ch	R/W	RGB_N_EQ3[7:0]								
R3Dh	R/W	RGB_CHGEN_ON[7:0]								
R3Eh	R/W	RGB_CHGEN_OFF[7:0]								
R3Fh	R/W	RGB_CHGEN_OFF2[7:0]								
Description	This command is used to set internal SD outptu timing.									
	Interlace_line : Set Even or Odd flag between SD and GIP.									
	RGB_JDT[2:0] : Set source dot inversion type at normal mode.									
	RGB_JDT[2:0]		Dot inversion type							
	000		1-dot							
	001		1+2-dot							
	010		2-dot							
	011		4-dot							
	100		Column							
	101		Zig-Zag							
	110		interlace							
	Others		Inhibit							
RGB_N_EQ1[7:0] : Set SD EQ1 time. (Pull GND)										
RGB_N_EQ1[7:0]		Clock cycle								
00h		0 timing clock								
01h		1 timing clock								
02h		2 timing clock								
:		1 timing clock /step								
FEh		254 timing clock								
FFh		255 timing clock								
Note: 1 timing clock = 4 multiple of Fosc.										
RGB_N_EQ2[7:0] : Set SD EQ2 time. (VCI or Hi-Z)										
RGB_N_EQ2[7:0]		Clock cycle								
00h		0 timing clock								
01h		1 timing clock								
02h		2 timing clock								
:		1 timing clock /step								
FEh		254 timing clock								
FFh		255 timing clock								
Note: 1 timing clock = 4 multiple of Fosc.										
RGB_N_EQ2_Temp[7:0] : Set SD EQ2 time for low temperature. (VCI or Hi-Z)										
RGB_N_EQ2[7:0]		Clock cycle								
00h		0 timing clock								
01h		1 timing clock								

JD9365D

02h	2 timing clock
:	1 timing clock /step
FEh	254 timing clock
FFh	255 timing clock

Note: 1 timing clock = 4 multiple of Fosc.

RGB _N_EQ3[7:0]: Set SD EQ3 time. (VCI or Hi-Z)

RGB _N_EQ3[7:0]	Clock cycle
00h	0 timing clock
01h	1 timing clock
02h	2 timing clock
:	1 timing clock /step
FEh	254 timing clock
FFh	255 timing clock

Note: 1 timing clock = 4 multiple of Fosc.

RGB _CHGEN_ON[7:0]: Set SD output time.

RGB _CHGEN_ON[7:0]	Clock cycle
00h	0 timing clock
01h	1 timing clock
02h	2 timing clock
:	1 timing clock /step
FEh	254 timing clock
FFh	255 timing clock

Note: 1 timing clock = 4 multiple of Fosc.

RGB _CHGEN_OFF[7:0]: Set 1st SD OP output time.

RGB _CHGEN_OFF[7:0]	Clock cycle
00h	0 timing clock
01h	1 timing clock
02h	2 timing clock
:	1 timing clock /step
FEh	254 timing clock
FFh	255 timing clock

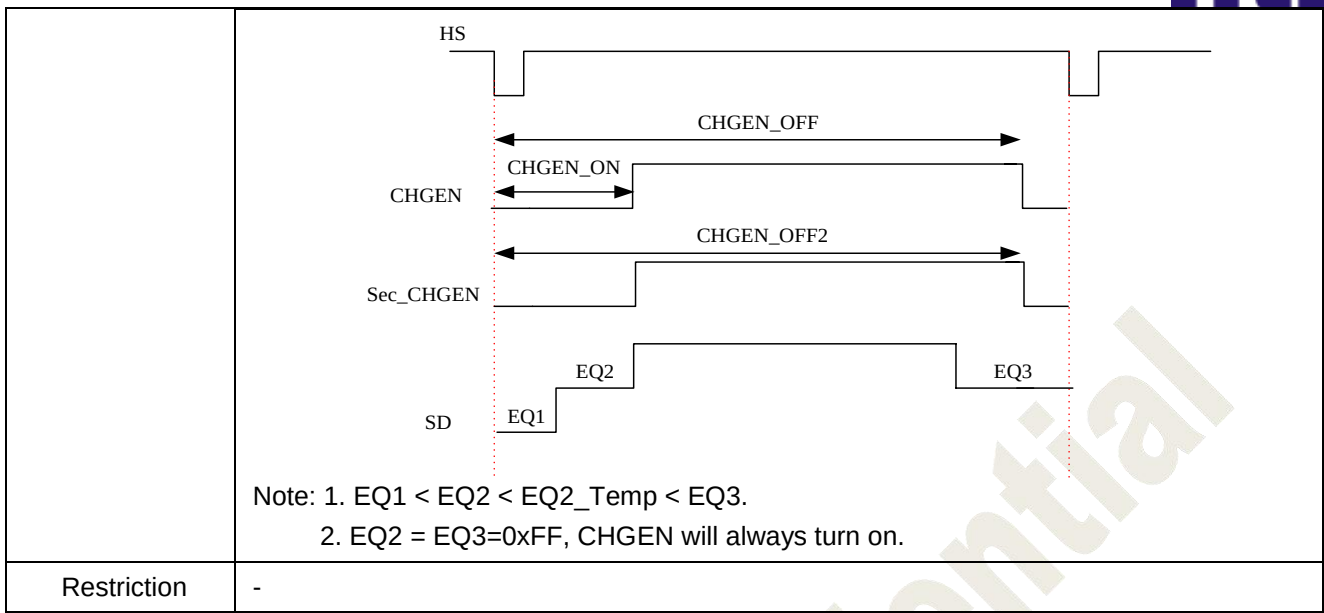
Note: 1 timing clock = 4 multiple of Fosc.

RGB _CHGEN_OFF2[7:0]: Set 2nd SD OP output time

RGB _CHGEN_OFF2[7:0]	Clock cycle
00h	0 timing clock
01h	1 timing clock
02h	2 timing clock
:	1 timing clock /step
FEh	254 timing clock
FFh	255 timing clock

Note: 1 timing clock = 4 multiple of Fosc.

JD9365D



JD9365D

2612 SET TCON: Timing control setting (Page1-R40h~R4Ch)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R40h	R/W	X	IP750_1	LN[1:0]		IP750_0	RSO[2:0]			
R41h	R/W	LN[9:2]								
R42h	R/W	SLT[7:0]								
R43h	R/W	VFP[7:0]								
R44h	R/W	VBP[7:0]								
R45h	R/W	HBP[7:0]								
R46h	R/W	TE_Delay[10:3]								
R47h	R/W	BlkinF_num[7:0]								
R48h	R/W	BlkinF_STR[7:0]								
R49h	R/W	BlkinFra me	X	X	X	X	TE_Delay[2:0]			
R4Ah	R/W	FRAME_NUM[1:0]		Auto_Te st_en	TEST_P AT_EN	TEST_PATTERN[3:0]				
R4Bh	R/W	TCON_OPT1[15:8]								
R4Ch	R/W	TCON_OPT1[7:0]								

This command is used to set timing control.

RSO[1:0]: Resolution select.

RSO[2:0]	Resolution	Source channels
000	Inhibit	Inhibit
001	750RGB x LN*2 dot	Refer to {IP750_1,IP750_0}
010	600RGBx LN*2 dot	S1~S900, S1503~S2400
011	640RGB x LN*2 dot	S1~S960, S1443~S2400
100	720RGB x LN*2 dot	S1 ~ S1080, S1323~S2400
101	768RGB x LN*2 dot	S1 ~ S1152, S1251~S2400
110	800RGB x LN*2 dot	S1 ~ S2400
111	Inhibit	Inhibit

RSO[2:0]	{IP750_1,IP750_0}	Source channels
001	00	S1~S1125, S1276~S2400
	01	S1~S1152, S1249~S2346
	10	S55~S1152, S1249~S2400

LN[9:0]: Sets the gate line number to drive LCD panel. The number of lines must be equal or more than the gate line number of LCD panel. Gate line number = LN*2.

NL[9:0]	Gate Line
---------	-----------

Description

0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	1	2
0	0	0	0	0	0	0	0	1	0	4
-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-
1	0	0	1	1	1	1	1	1	1	1278
1	0	1	0	0	0	0	0	0	0	1280
1	0	1	0	0	0	0	0	0	1	1282
-	-	-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-	-	-
1	1	1	1	1	1	1	1	0	1	2042
1	1	1	1	1	1	1	1	1	0	2044
1	1	1	1	1	1	1	1	1	1	2046

SLT[7:0]: Sets the scan line time width. Scan line time width = osc*4.

STL[7:0]								Scan Line time(osc)
0	0	0	0	0	0	0	0	Inhibit
0	0	0	0	0	0	0	1	4
0	0	0	0	0	0	1	0	8
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
0	1	1	1	1	1	1	1	508
1	0	0	0	0	0	0	0	512
1	0	0	0	0	0	0	1	516
-	-	-	-	-	-	-	-	-
-	-	-	-	-	-	-	-	-
1	1	1	1	1	1	0	1	1012
1	1	1	1	1	1	1	0	1016
1	1	1	1	1	1	1	1	1020

VBP[7:0]: Vertical back porch number setting.

VFP[7:0]: Vertical front porch number setting.

HBP[7:0]: Horizontal front porch number setting.

VBP[7:0] / VFP[7:0] / HBP[7:0]	Number of VBP / VFP (line)	Number of HBP(PCLK)
0x00	Inhibit	Inhibit
0x01	1	1

0x02	2	2
:	:	:
0xFE	254	254
0xFF	255	255

Note: VFP/VBP set 4 line unless, HBP set 2 PCLK unless.

TE_Delay[7:0]: TE delay setting.

TEST_PAT_EN: AGING mode enable/ disable, 1=>Enable, 0=>Disable.

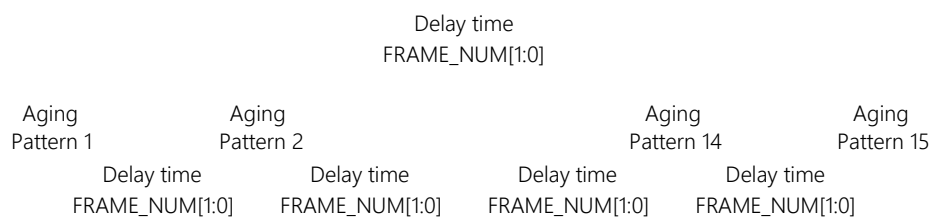
Auto_Test_en: Automatically executed aging cycle. 1=>Enable, 0=>Disable(display data by PATTERN_SELECT[3:0]).

Test Flow:

Step 1: Send slpout command

Step 2: Set Auto_Test_EN=1 & Test_PAT_EN=1.

Internal pattern will run differential pattern automatically.



FRAME_NUM[1:0]: Interval of each pattern when Automatically executed aging cycle.

JD9365D

TEST_PATTERN[3:0]: Display pattern select when AGING_EN=1 and AUTO_EN=0.

TE_Delay[10:0]	TE output position (line)
0x000	VFP start position
0x001	1 gate line
0x002	2 gate line
:	:
0x500	1280 gate line
Others	Inhibit



FRAME_NUM[1:0]		Frame number
0	0	32
0	1	96
1	0	160

PATTERN_SELECT[3:0]				Display pattern
0	0	0	0	Full Black
0	0	0	1	Full White
0	0	1	0	Full Red
0	0	1	1	Ful Green
0	1	0	0	Full Blue
0	1	0	1	Color bar(H)
0	1	1	0	Grayscale(V)
0	1	1	1	Grayscale(H)
1	0	0	0	Pixel 64x64
1	0	0	1	Crosstalk with G0
1	0	1	0	Crosstalk with G255
1	0	1	1	Flciker pattern for column inversion
1	1	0	0	Flicker pattern for 1 dot inversion
1	1	0	1	Flicker pattern for 2 dot inversion
1	1	1	0	Parity complement(for CP test)
1	1	1	1	Border(external : red, middle: blue internal: black)

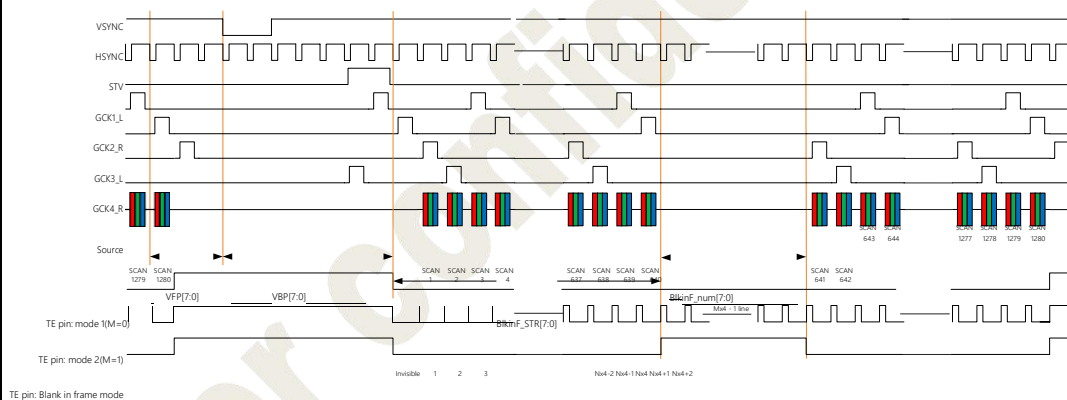
BlkinFrame: Blking in Frame function, set this bit 1 to enable this function.

BlkinF_num [7:0]: Blking in Frame output pulse width. Only use in BlkinFrame=1.

BlkinF_num[7:0]	Blking pulse width (line)
8h'00	Inhibit
8h'01	1x4 - 1 = 3 line
8h'02	2x4 - 1 = 7 line
8h'03	3x4 - 1 = 11 line
:	:
N	Mx4 - 1 line
N+1	(M+1)x4 - 1 line
:	:
8h'FD	253x4 - 1 = 1011 line
8h'FE	254x4 - 1 = 1015 line
8h'FF	255x4 - 1 = 1019 line

BlkinF_STR [7:0]: Blking in Frame output start position. Only use in BlkinFrame=1.

BlkinF_num[7:0]	Blking pulse start position (line)
8h'00	Inhibit
8h'01	Invisible + (1x4 +1) line
8h'02	Invisible + (2x4 +1) line
8h'03	Invisible + (3x4 +1) line
:	:
N	Invisible + (Nx4 +1) line
N+1	Invisible + ((N+1)x4 +1) line
:	:
8h'FD	Invisible + (253x4 +1) line
8h'FE	Invisible + (254x4 +1) line
8h'FF	Invisible + (255x4 +1) line



TCON_OPT1[15:8]: TCON option.

TCON_OPT1[7:0]: TCON option.

TCON_OPT [10]:Source EQ off. 0: Source EQ on, 1: Source EQ off.

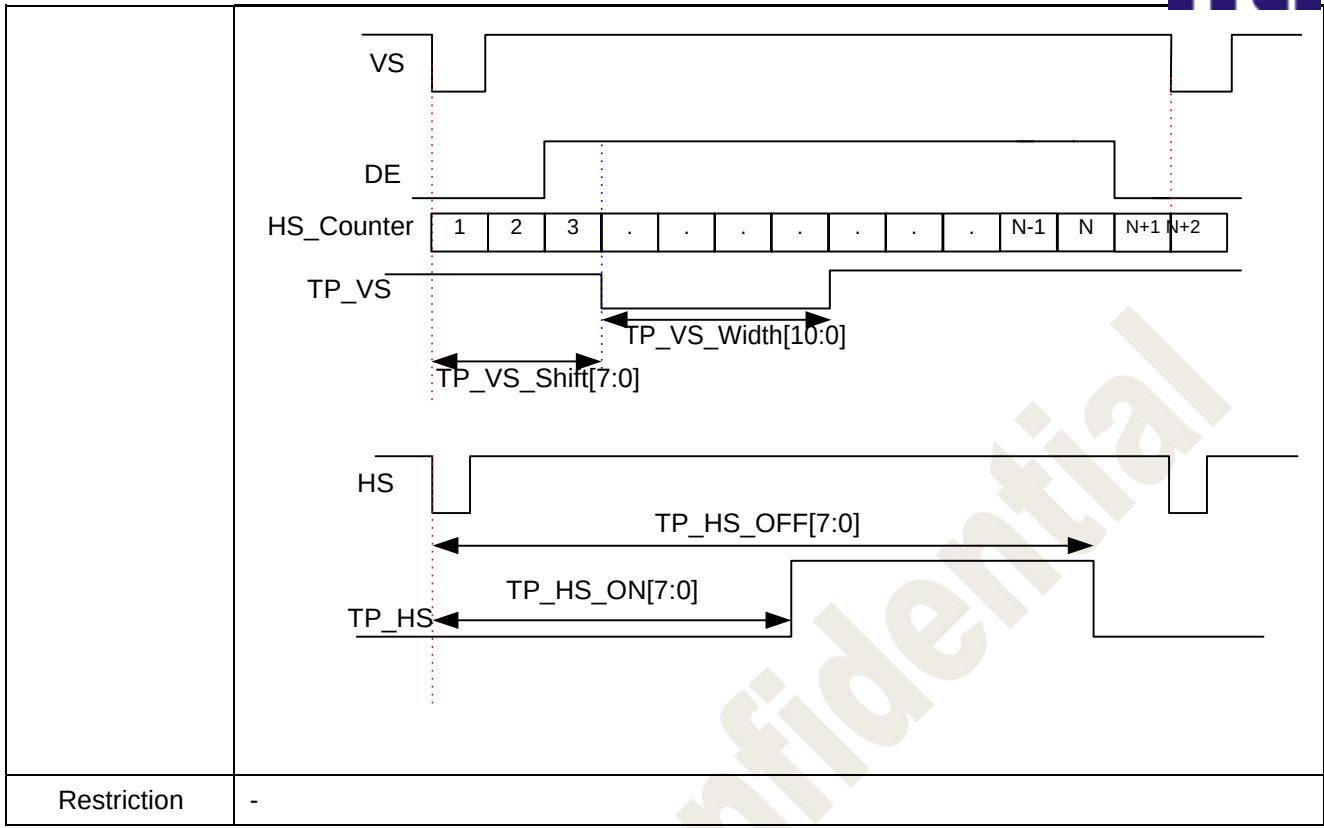
Restriction

-

JD9365D

2613. SET TE PAD: TE pad function setting (Page1-R4Dh~R53h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R4Dh	R/W	X	X	X	X	X	X	X	X	
R4Eh	R/W	X	X	X	X	TP_VS_EN	TP_VS_WIDTH[10:8]			
R4Fh	R/W	TP_VS_WIDTH[7:0]								
R50h	R/W	TP_VS_SHIFT[7:0]								
R51h	R/W	X	X	X	X	X	X	TP_HS_BLK_EN	TP_HS_DIS_EN	
R52h	R/W	TP_HS_ON[7:0]								
R53h	R/W	TP_HS_OFF[7:0]								
Description	This command is used to set TE and TE1 output state. TP_VS_EN: Set VS outptu enable/Disable. 0: When TE/TE1=10, TE/TE1 will outptu at Low. 1: Normal output TP_VS_WIDTH[10:0]: Set width of output low. TP_VS_SHIFT[7:0]: Set width of output high. TP_VS_WIDTH[10:0]=0 is inhibited. TP_VS_WIDTH[10:0]+ TP_VS_SHIFT[7:0]<VFP+VS+VBP+DISP AREA TP_HS_BLK_EN: Set HS output enable/disable at blanking area. TP_HS_DIS_EN: Set HS output enable/disable at disaply area. TP_HS_ON[7:0]: Set HS output low time. TP_HS_OFF[7:0]: Set HS output high time. TP_HS_ON[7:0] must small than TP_HS_OFF[7:0] Related diagram chart as below:									



JD9365D

2614. POWER_STATE: Power status setting (Page1-R54h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R54h	R/W	-	-	-	-	-	-	-	DSTBY	
Description	This command is used to set power state. DSTBY: Standby mode select. DSTBY=1, driver IC enters the deep standby mode when POW_EN=0. DSTBY=0, driver IC enters the standby mode when POW_EN=0. When DSTBY=1, all the display operation stops, All the internal operations and R-C oscillators are suspended. Driver IC should receive reset pull low 3ms at least to exist deep standby mode.									
Restriction	-									

JD9365D

2615. DCDC_SEL: Power Mode and Charge Pump Setting (Page1-R55h~ R5Ch)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R55h	R/W	X	X	X	X	DCDCM[3:0]				
R56h	R/W	X	X	X	X	AUTO_RT	X	AVDD_RT[1:0]		
R57h	R/W	VGH_RT[2:0]			VGL_RT[2:0]			VCL_RT[1:0]		
R58h	R/W	X	X	X	AVDD[4:0]					
R59h	R/W	VCL[2:0]			AVEE[4:0]					
R5Ah	R/W	X	VGH[6:0]							
R5Bh	R/W	X	X	VGL[5:0]						
R5Ch	R/W	X	X	VDDA_CLK[1:0]		VCL_CLK[1:0]		VGHVGL_CLK[1:0]		

Description

This command is used to Trimming HS Data Lane 0/1 Delay.

DCDCM[3:0]: The bits just valid under BOOSTM[1:0]=00

DCDCM[3:0]	AVDD	AVEE	VCL
0000	Internal Charge Pump(CP)		
0001	JD5001		CP
0010	NT power IC		CP
0011	HX power IC		CP
0100	PFM mode 1		CP
0101	PFM mode 2		CP
0110	PFM mode 3		CP
0111	inhibit		
1000	No output		CP
1001	JD5001		
1010	NT power IC		
1011	No output		CP
1100	AVDD/AVEE external when SLPOUT		CP
1101	AVDD external when SLPOUT		CP
1110	AVEE external when SLPOUT		CP
1111	No output		CP

Note:

1. The connection between JD power IC / PFM mode1~3 and JD9311 could refer PXXX.
2. CP: Dr.IC Internal charge pump.

AUTO_RT: Auto pumping ratio function enable. When auto pump function is turn on, charge pumping circuit will detect voltage of VCI to select suitable ratio for charge pump circuit.

AUTO_RT=0, turn off auto pump.

AUTO_RT=1, turn on auto pump.

AVDD_RT[1:0]: Specify the charge pump ratio of AVDD under BOOSTM[1:0]=00.(Internal charge pump)

AVDD_RT[1:0]	Charge pump Ratio
0h	1.5 x VCIP
1h	2.0 x VCIP
2h	2.0 x VCIP
3h	3.0 x VCIP

VGH_RT[2:0]: Specify the charge pump ratio of VGH.

VGL_RT[2:0]: Specify the charge pump ratio of VGL.

VGH_RT[2:0]	Charge pump Ratio	VGL_RT[2:0]	Charge pump Ratio
0h	AVDD-VCL	0h	VCL-AVDD
1h	2*AVDD	1h	AVEE-AVDD
2h	2*AVDD-VCL	2h	AVEE+VCL-AVDD
3h	2*AVDD-AVEE	3h	2*AVEE-AVDD
4h	3*AVDD-VCL	4h	inhibit
5h	3*AVDD-AVEE	5h	inhibit
6h	3*AVDD-AVEE-VCL	6h	inhibit
7h	3*AVDD-2*AVEE	7h	inhibit

Note: If VGH_RT[2:0]=100, VGL_RT[1:0] can only be set 01 or 11.

VCL_RT[1:0]: Specify the charge pump ratio of VGL.

VCL_RT[1:0]	Charge pump Ratio
0h	inhibit
1h	- VCIP
2h	inhibit
3h	inhibit

AVDD_S[4:0]: Set clamping voltage level of AVDD.

AVEE_S[4:0]: Set clamping voltage level of AVEE.

AVDD_S[4:0]	AVDD (V)	AVEE_S[4:0]	AVEE (V)
0x00	6.5	0x00	-6.5
0x01	6.4	0x01	-6.4
⋮	100mv / step	⋮	100mv / step
0x1E	3.5	0x1E	-3.5
0x1F	3.4	0x1F	-3.4

VCL_S[1:0]: Set clamping voltage level of VCL.

VCL_S[1:0]	VCL (V)	VCL_S[2:0]	VCL (V)
0x00	-2.5	0x04	-3.3
0x01	-2.7	0x05	-3.5
0x02	-2.9	0x06	-3.7
0x03	-3.1	0x07	-3.9

VGH_S[6:0]: Set clamping voltage level of VGH.

VGH_S[6]	VGH_S[5:0]	VGH (V)
0	00h	7.0
0	01h	7.2
0	02h	7.4
0	⋮	200mv / step
0	3Dh	19.2
0	3Eh	19.4
0	3Fh	19.6
1	00h	7.8
1	01h	8.0
1	02h	8.2
1	⋮	200mv / step

JD9365D

	.	
1	3Dh	20.0
1	3Eh	20.2
1	3Fh	20.4

VGL_S[5:0]: Set clamping voltage level of VGL.

VGL_S[5:0]	VGL (V)
0	-7.0
1	-7.2
2	-7.4
.	200mv / step
.	
.	
2Eh	-15.6
2Fh	-15.8
2Dh	-16.0
Others	Inhibit

VGHVGL_CLK[1:0]: Set frequency of pumping clock of VGH/VGL under normal mode.

VGHVGL_CLK[1:0]	Ratio
0x00	HSCLK / 4
0x01	HSCLK / 2
0x02	HSCLK
0x03	HSCLK x 2

Note: HSCLK is define frequency of 1-line.

VCL_CLK[1:0]: Set frequency of pumping clock of VCL under normal mode.

VCL_CLK[1:0]	Ratio
0x00	HSCLK / 4
0x01	HSCLK / 2
0x02	HSCLK
0x03	HSCLK x 2

Note: HSCLK is define frequency of 1-line.

VDVA_CLK[1:0]: Set frequency of pumping clock of VDVA under normal mode.

VCL_CLK[1:0]	Ratio
0x00	HSCLK / 4
0x01	HSCLK / 2
0x02	HSCLK
0x03	HSCLK x 2

Note: HSCLK is define frequency of 1-line.

Restriction

-

JD9365D

2616. SET_GAMMA: Set Gamma output voltage (Page1-R5Dh~ R82h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R5Dh	R/W	X	RPA18[6:0]							
R5Eh	R/W	X	RPA17[6:0]							
R5Fh	R/W	X	RPA16[6:0]							
R60h	R/W	X	RPA15[6:0]							
R61h	R/W	X	RPA14[6:0]							
R62h	R/W	X	RPA13[6:0]							
R63h	R/W	X	RPA12[6:0]							
R64h	R/W	X	RPA11[6:0]							
R65h	R/W	X	RPA10[6:0]							
R66h	R/W	X	RPA9[6:0]							
R67h	R/W	X	RPA8[6:0]							
R68h	R/W	X	RPA7[6:0]							
R69h	R/W	X	RPA6[6:0]							
R6Ah	R/W	X	RPA5[6:0]							
R6Bh	R/W	X	RPA4[6:0]							
R6Ch	R/W	X	RPA3[6:0]							
R6Dh	R/W	X	RPA2[6:0]							
R6Eh	R/W	X	RPA1[6:0]							
R6Fh	R/W	X	RPA0[6:0]							
R70h	R/W	X	RNA18[6:0]							
R71h	R/W	X	RNA17[6:0]							
R72h	R/W	X	RNA16[6:0]							
R73h	R/W	X	RNA15[6:0]							
R74h	R/W	X	RNA14[6:0]							
R75h	R/W	X	RNA13[6:0]							
R76h	R/W	X	RNA12[6:0]							
R77h	R/W	X	RNA11[6:0]							
R78h	R/W	X	RNA10[6:0]							
R79h	R/W	X	RNA9[6:0]							
R7Ah	R/W	X	RNA8[6:0]							
R7Bh	R/W	X	RNA7[6:0]							
R7Ch	R/W	X	RNA6[6:0]							
R7Dh	R/W	X	RNA5[6:0]							
R7Eh	R/W	X	RNA4[6:0]							

JD9365D

R7Fh	R/W	X	RNA3[6:0]	
R80h	R/W	X	RNA2[6:0]	
R81h	R/W	X	RNA1[6:0]	
R82h	R/W	X	RNA0[6:0]	
Description	This command is used to set postive /neagative volatge of source output. The related illustration can refer "8.1 Grayscale-Level adjustment control" of JD9365 Datasheet.			
Restriction	-			

fitipower confidential

JD9365D

2617. SET DDB: Set Device Descriptor Block (Page1-R83h~R87h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R83h	R	X	X	X	X	X	X	X	X	
R84h	R/W	DDB1[7:0]								
R85h	R/W	DDB2[7:0]								
R86h	R/W	DDB3[7:0]								
R87h	R/W	DDB4[7:0]								
Description	This command is used to set DDB value and read form A1h and A8h.									
Restriction	-									

JD9365D

2618. SETECO(Page1-R88h~ R8Ch)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
R88h	R	-	-	-	-	-	-	-	-	
R89h	R/W	ECO1[7:0]								
R8Ah	R/W	ECO2[7:0]								
R8Bh	R/W	ECO3[7:0]								
R8Ch	R/W	ECO4[7:0]								
Description	This command is used to set ECO. ECO1~4[7:0]: Internal option.									
Restriction	-									

JD9365D

2619. SET_TE_PAD2(Page1-RAFh~ RB1h)

CMD_ADD	R/W	D7	D6	D5	D4	D3	D2	D1	D0	HEX
RAFh	R/W	-	-	-	VHS_ TEON_ OPT	TP_VS_ EN2	TP_VS_WIDTH2[10:8]			
RB0h	R/W	TP_VS_WIDTH2[7:0]								
RB1h	R/W	TP_VS_SHIFT2[7:0]								
Description	VHS_TEON_OPT: 1'b0:TP_VS and TP_HS don't use TEON 1'b1:TP_VS and TP_HS use TEON TP_VS_EN2: Set 2nd TE_VS output enable/Disable. This function can be used when TP_VS_EN=1. 0: When TE/TE1=10, TE/TE1 2nd VS will output at Low. 1: Normal output TP_VS_WIDTH2[10:0]: Set width2 of output low. TP_VS_SHIFT2[7:0]: Set width2 of output high.									
Restriction	-									

3. OTP Programming

3.1. OTP Table

OTP_ INDEX (HEX)	Reg. Def.	Page	Reg. Add.	Duplicate	B7	B6	B5	B4	B3	B2	B1	B0
0	SEQUENCE_CTL	P0	70	1	NVALID	DC0H[2:0]			-	DC1H[2:0]		
1			71		-	DC2H[2:0]			-	DC3H[2:0]		
2			72		-	-	-	-	DC7H[2:0]			
3			73		DC0L	DC1L	DC2L	DC3L	DC4L	DC5L	DC6L	DC7L
4			74		T0P[1:0]		T1P[1:0]		T8P[1:0]		-	-
5			75		-	SOFT0H[2:0]			-	SOFT0L[2:0]		
6	SETID(OTP*5)	P0	77	1	NVALID	-	-	-	-	-	-	-
7			78		ID1[7:0]							
8			79		ID2[7:0]							
9			7A		ID3[7:0]							
A			7B		ID4[7:0]							
B			77	2	NVALID	-	-	-	-	-	-	-
C			78		ID1[7:0]							
D			79		ID2[7:0]							
E			7A		ID3[7:0]							
F			7B		ID4[7:0]							
10			77	3	NVALID	-	-	-	-	-	-	-
11			78		ID1[7:0]							
12			79		ID2[7:0]							
13			7A		ID3[7:0]							
14			7B		ID4[7:0]							
15			77	4	NVALID	-	-	-	-	-	-	-
16			78		ID1[7:0]							
17			79		ID2[7:0]							
18			7A		ID3[7:0]							
19			7B		ID4[7:0]							
1A			77	5	NVALID	-	-	-	-	-	-	-
1B			78		ID1[7:0]							
1C			79		ID2[7:0]							
1D			7A		ID3[7:0]							
1E			7B		ID4[7:0]							
1F	SETDSI	P0	80	1	NVALID	DSI_INIT0[6:0]						
20	SETUP		81		DSI_INIT1[7:0]							

21			82		DSI_INIT2[7:0]						
22	SETDSI SETUP2	P0	83	1	NVALID	DSI_INIT3[6:0]					
23	SET_MT K	P0	BE	1	NVALID	-	-	-	-	-	MASK_V S_REV ITLCE_F RM
24	SET_WD	PA	E6	1	NVALID	-	-	-	-	-	WD_MODE[1:0]
25			E7		-	-	-	WD_off	WD_Timer[3:0]		
26	VCOM_ SET(OT P*5)	P1	00	1	NVALID	-	-	-	-	-	VCOM_S[8]
27			01		VCOM_S[7:0]						
28			00	2	NVALID	-	-	-	-	-	VCOM_S[8]
29			01		VCOM_S[7:0]						
2A			00	3	NVALID	-	-	-	-	-	VCOM_S[8]
2B			01		VCOM_S[7:0]						
2C			00	4	NVALID	-	-	-	-	-	VCOM_S[8]
2D			01		VCOM_S[7:0]						
2E			00	5	NVALID	-	-	-	-	-	VCOM_S[8]
2F			01		VCOM_S[7:0]						
30	VCOM_ R_SET(OTP*5)	P1	03	1	NVALID	-	-	-	-	-	VCOM_R _S[8]
31			04		VCOM_R_S[7:0]						
32			03	2	NVALID	-	-	-	-	-	VCOM_R _S[8]
33			04		VCOM_R_S[7:0]						
34			03	3	NVALID	-	-	-	-	-	VCOM_R _S[8]
35			04		VCOM_R_S[7:0]						
36			03	4	NVALID	-	-	-	-	-	VCOM_R _S[8]
37			04		VCOM_R_S[7:0]						
38			03	5	NVALID	-	-	-	-	-	VCOM_R _S[8]
39			04		VCOM_R_S[7:0]						
3A	PFM_CT RL	P1	06	1	NVALID	-	-	-	PFM_CLK[3:0]		
3B			07		DUTYPS[3:0]			DUTYNS[3:0]			
3C			08		DUTYP[3:0]			DUTYN[3:0]			
3D			09		PFM_PDS[2:0]			PFM_PD[2:0]			
3E	PWRIC_ SET	P1	0A	1	NVALID	-	JD_RELD	VCSW_S WAP	EN_VREF_S[3:0]		
3F			0B		PWRIC_SET[7:0]						
40			0C		-	PWRIC_CLK1[2:0]		-	PWRIC_IDLE_CLK1[2:0]		

41			0D		HX_SOFT	PWRIC_CLK2[2:0]			PWRIC_HXCLK[3:0]			
42			0E		-	-	-	-	-	-	LED_PWR_S[1:0]	
4B	GAMMA_SET	P1	17	1	NVALID	-	-	VGMP_S[8]	-	-	-	VGSP_S[8]
4C			18		VGMP_S[7:0]							
4D			19		VGSP_S[7:0]							
4E			1A		-	-	-	VGMP_S[8]	-	-	-	VGSN_S[8]
4F			1B		VGMP_S[7:0]							
50			1C		VGSN_S[7:0]							
51	OTP_SE T	P1	1D	1	NVALID	-	VPP_DT[1:0]		-	VPP_VGH_RT[2:0]		
52			1E		-	VPP_VGH_S[6:0]						
53	GATE_P O W E R	P1	1F	1	NVALID	VGH_REG_S[6:0]						
54			20		-	-	VGL_REG_S[5:0]					
55			21		-	-	VGL_REG2_S[5:0]					
56			22		-	VGH_RE G_SHT	VGL_RE G_SHT	VGL_RE G2_SHT	VGH_RE G_EN	VGL_RE G_EN	VGL_RE G2_EN	SHT_VG L_REG
57	DCDC_C TRL	P1	23	1	NVALID	-	-	-	VGH_NC	VGL_NC	VCLREG_OP_CEN	VCL_NC
58			24		VGHEN	VGLEN	AVDDEN	AVEEEN	VCLEN	VDDAEN	VCLREG_EN	VCL2VD DN
59	POWAM P_CTRL	P1	25	1	NVALID	AP[2:0]			-	-	-	-
5A			26		GAS_EN	VCOM_EN	GAS_VC OM_EN	VGM_EN	GAS_AV DD_EN	GAS_AV EE_EN	EN_VDD D_VCI	EN_VDDD _IOVCC
5B			27		UP_VDD _1P8_EN	VDDD_S1[2:0]			-	VDDD_S2[2:0]		
68	SETSTB A	P1	35	1	NVALID	GAP[2:0]			SAP[3:0]			
69			36		-	-	-	-	SAP2[3:0]			
6A	SETPAN EL	P1	37	1	NVALID	Z_line	ENZ[1:0]		SS_PAN EL	GS_PAN EL	REV_PA NEL	BGR_PA NEL
6B	SETRGB CYC	P1	38	1	NVALID	interlace_line	-	-	-	RGB_JDT[2:0]		
6C			39		RGB_N_EQ1[7:0]							
6D			3A		RGB_N_EQ2[7:0]							
6E			3B		RGB_N_EQ2_Temp[7:0]							
6F			3C		RGB_N_EQ3[7:0]							
70			3D		RGB_CHGEN_ON[7:0]							
71			3E		RGB_CHGEN_OFF[7:0]							
72			3F		RGB_CHGEN_OFF2[7:0]							
73	SET_TC ON	P1	40	1	NVALID	IP750_1	LN[1:0]		IP750_0	RSO[2:0]		
74			41		LN[9:2]							
75			42		SLT[7:0]							

76			43	1	VFP[7:0]							
77			44		VBP[7:0]							
78			45		HBP[7:0]							
79			46		TE_Delay[10:3]							
7A			47		BlkinF_num[7:0]							
7B			48		BlkinF_STR[7:0]							
7C			49		BlkinFrame	BlkinF_hsync	-	-	-	TE_Delay[2:0]		
7D			4A		FRAME_NUM[1:0]		Auto_Test_en	TEST_PAT_T_EN	TEST_PATTERN[3:0]			
7E		4B	TCON_OPT1[15:8]									
7F		4C	TCON_OPT1[7:0]									
80	SET_TE_PAD	P1	4D	1	NVALID	-	-	-	-	-	-	-
81			4E		-	-	-	-	TP_VS_EN	TP_VS_WIDTH[10:8]		
82			4F		TP_VS_WIDTH[7:0]							
83			50		TP_VS_SHIFT[7:0]							
84			51		-	-	-	-	-	TP_HS_BLK_EN	TP_HS_DIS_EN	
85			52		TP_HS_ON[7:0]							
86			53		TP_HS_OFF[7:0]							
87	DCDC_SEL	P1	55	1	NVALID	-	-	-	DCDCM[3:0]			
88			56		-	-	-	-	AUTO_RT	-	AVDD_RT[1:0]	
89			57		VGH_RT[2:0]			VGL_RT[2:0]			VCL_RT[1:0]	
8A			58		-	-	-	AVDD_S[4:0]				
8B			59		VCL_S[2:0]			AVEE_S[4:0]				
8C			5A		-	VGH_S[6:0]						
8D			5B		-	-	VGL_S[5:0]					
8E			5C		-	-	VDDA_CLK[1:0]		VCL_CLK[1:0]		VGHVGL_CLK[1:0]	
8F	SETDDB	P1	83	1	NVALID	-	-	-	-	-	-	-
90			84		DDB1[7:0]							
91			85		DDB2[7:0]							
92			86		DDB3[7:0]							
93			87		DDB4[7:0]							
CD	SETCUSTM_ID	P1	A6	1	NVALID	-	-	-	-	-	-	-
CE			A7		CUSTM1[7:0]							
CF			A8		CUSTM2[7:0]							
D0			A9		CUSTM3[7:0]							
D1			AA		CUSTM4[7:0]							
D2			AB		CUSTM5[7:0]							

Preliminary

JD9365D

D3			AC		CUSTM6[7:0]					
D4			AD		CUSTM7[7:0]					
D5			AE		CUSTM8[7:0]					
D6	SET_TE _PAD2	P1	AF	1	NVALID	-	-	VHS_TE	TP_VS_E	TP_VS_WIDTH2[10:8]
D7			B0		TP_VS_WIDTH2[7:0]					
D8			B1		TP_VS_SHIFT2[7:0]					
100	SET_GA MMA(OT P*3)	P1	5D	1	NVALID	RPA18[6:0]				
101			5E		-	RPA17[6:0]				
102			5F		-	RPA16[6:0]				
103			60		-	RPA15[6:0]				
104			61		-	RPA14[6:0]				
105			62		-	RPA13[6:0]				
106			63		-	RPA12[6:0]				
107			64		-	RPA11[6:0]				
108			65		-	RPA10[6:0]				
109			66		-	RPA9[6:0]				
10A			67		-	RPA8[6:0]				
10B			68		-	RPA7[6:0]				
10C			69		-	RPA6[6:0]				
10D			6A		-	RPA5[6:0]				

10E		6B	-	RPA4[6:0]
10F		6C	-	RPA3[6:0]
110		6D	-	RPA2[6:0]
111		6E	-	RPA1[6:0]
112		6F	-	RPA0[6:0]
113		70	-	RNA18[6:0]
114		71	-	RNA17[6:0]
115		72	-	RNA16[6:0]
116		73	-	RNA15[6:0]
117		74	-	RNA14[6:0]
118		75	-	RNA13[6:0]
119		76	-	RNA12[6:0]
11A		77	-	RNA11[6:0]
11B		78	-	RNA10[6:0]
11C		79	-	RNA9[6:0]
11D		7A	-	RNA8[6:0]
11E		7B	-	RNA7[6:0]
11F		7C	-	RNA6[6:0]
120		7D	-	RNA5[6:0]
121		7E	-	RNA4[6:0]
122		7F	-	RNA3[6:0]
123		80	-	RNA2[6:0]
124		81	-	RNA1[6:0]
125		82	-	RNA0[6:0]
126		5D	INVALID	RPA18[6:0]
127		5E	-	RPA17[6:0]
128		5F	-	RPA16[6:0]
129		60	-	RPA15[6:0]
12A		61	-	RPA14[6:0]
12B		62	-	RPA13[6:0]
12C		63	-	RPA12[6:0]
12D		64	-	RPA11[6:0]
12E		65	-	RPA10[6:0]
12F		66	-	RPA9[6:0]
130		67	-	RPA8[6:0]
131		68	-	RPA7[6:0]
132		69	-	RPA6[6:0]
133		6A	-	RPA5[6:0]
134		6B	-	RPA4[6:0]

135		6C	-	RPA3[6:0]
136		6D	-	RPA2[6:0]
137		6E	-	RPA1[6:0]
138		6F	-	RPA0[6:0]
139		70	-	RNA18[6:0]
13A		71	-	RNA17[6:0]
13B		72	-	RNA16[6:0]
13C		73	-	RNA15[6:0]
13D		74	-	RNA14[6:0]
13E		75	-	RNA13[6:0]
13F		76	-	RNA12[6:0]
140		77	-	RNA11[6:0]
141		78	-	RNA10[6:0]
142		79	-	RNA9[6:0]
143		7A	-	RNA8[6:0]
144		7B	-	RNA7[6:0]
145		7C	-	RNA6[6:0]
146		7D	-	RNA5[6:0]
147		7E	-	RNA4[6:0]
148		7F	-	RNA3[6:0]
149		80	-	RNA2[6:0]
14A		81	-	RNA1[6:0]
14B		82	-	RNA0[6:0]
14C		5D	INVALID	RPA18[6:0]
14D		5E	-	RPA17[6:0]
14E		5F	-	RPA16[6:0]
14F		60	-	RPA15[6:0]
150		61	-	RPA14[6:0]
151		62	-	RPA13[6:0]
152		63	-	RPA12[6:0]
153		64	-	RPA11[6:0]
154		65	-	RPA10[6:0]
155		66	-	RPA9[6:0]
156		67	-	RPA8[6:0]
157		68	-	RPA7[6:0]
158		69	-	RPA6[6:0]
159		6A	-	RPA5[6:0]
15A		6B	-	RPA4[6:0]
15B		6C	-	RPA3[6:0]

3

15C			6D	1	-	RPA2[6:0]		
15D			6E		-	RPA1[6:0]		
15E			6F		-	RPA0[6:0]		
15F			70		-	RNA18[6:0]		
160			71		-	RNA17[6:0]		
161			72		-	RNA16[6:0]		
162			73		-	RNA15[6:0]		
163			74		-	RNA14[6:0]		
164			75		-	RNA13[6:0]		
165			76		-	RNA12[6:0]		
166			77		-	RNA11[6:0]		
167			78		-	RNA10[6:0]		
168			79		-	RNA9[6:0]		
169			7A		-	RNA8[6:0]		
16A			7B		-	RNA7[6:0]		
16B			7C		-	RNA6[6:0]		
16C			7D		-	RNA5[6:0]		
16D			7E		-	RNA4[6:0]		
16E			7F		-	RNA3[6:0]		
16F			80		-	RNA2[6:0]		
170	81	-	RNA1[6:0]					
171	82	-	RNA0[6:0]					
172	SET_GI P_L	P2	00	1	INVALID	CGO1_PH_L	CGO1_INV_L	CGOUT1_L[4:0]
173			01		-	CGO2_PH_L	CGO2_INV_L	CGOUT2_L[4:0]
174			02		-	CGO3_PH_L	CGO3_INV_L	CGOUT3_L[4:0]
175			03		-	CGO4_PH_L	CGO4_INV_L	CGOUT4_L[4:0]
176			04		-	CGO5_PH_L	CGO5_INV_L	CGOUT5_L[4:0]
177			05		-	CGO6_PH_L	CGO6_INV_L	CGOUT6_L[4:0]
178			06		-	CGO7_PH_L	CGO7_INV_L	CGOUT7_L[4:0]
179			07		-	CGO8_PH_L	CGO8_INV_L	CGOUT8_L[4:0]
17A			08		-	CGO9_PH_L	CGO9_INV_L	CGOUT9_L[4:0]
17B			09		-	CGO10_PH_L	CGO10_INV_L	CGOUT10_L[4:0]
17C			0A		-	CGO11_PH_L	CGO11_INV_L	CGOUT11_L[4:0]
17D			0B		-	CGO12_PH_L	CGO12_INV_L	CGOUT12_L[4:0]
17E			0C		-	CGO13_PH_L	CGO13_INV_L	CGOUT13_L[4:0]
17F			0D		-	CGO14_PH_L	CGO14_INV_L	CGOUT14_L[4:0]
180			0E		-	CGO15_PH_L	CGO15_INV_L	CGOUT15_L[4:0]
181			0F		-	CGO16_PH_L	CGO16_INV_L	CGOUT16_L[4:0]
182			10		-	CGO17_PH_L	CGO17_INV_L	CGOUT17_L[4:0]

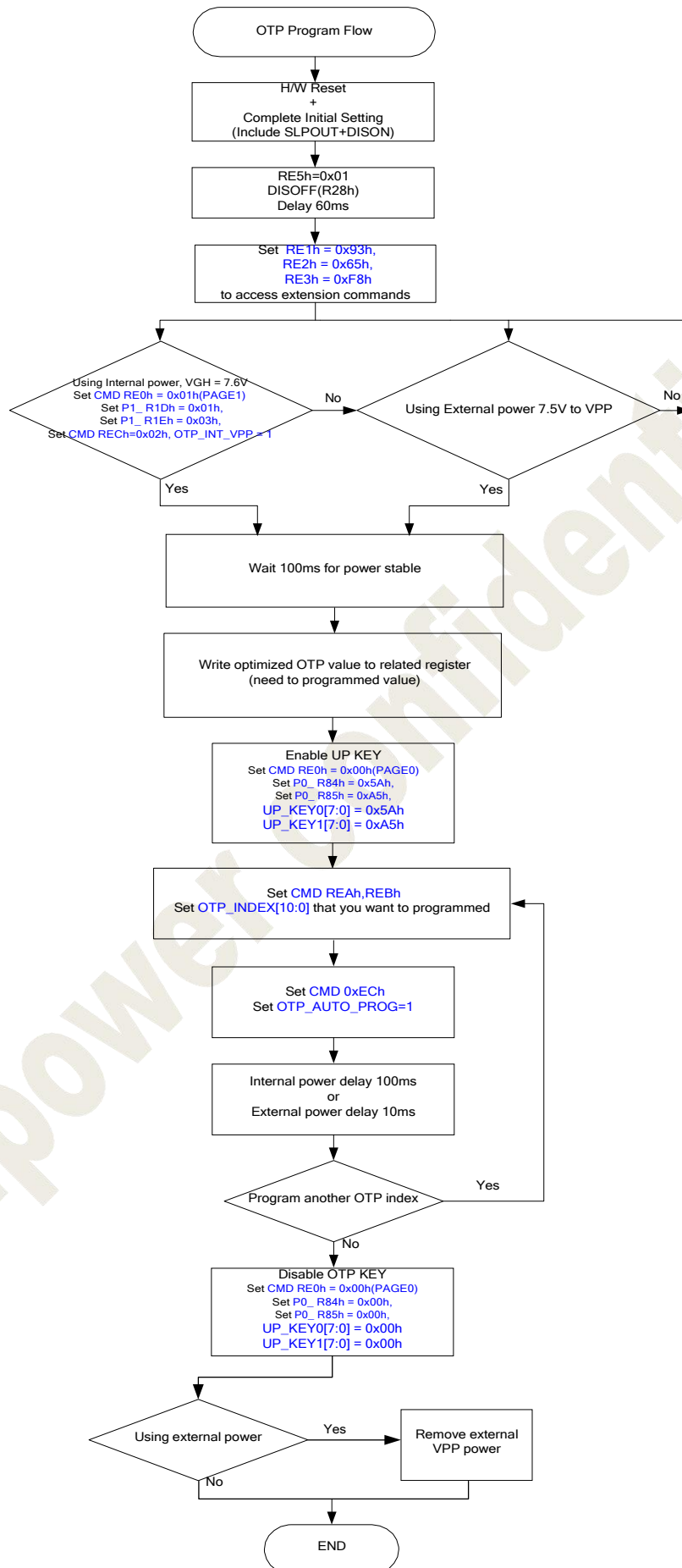
183			11		-	CGO18_PH_L	CGO18_INV_L	CGOUT18_L[4:0]
184			12		-	CGO19_PH_L	CGO19_INV_L	CGOUT19_L[4:0]
185			13		-	CGO20_PH_L	CGO20_INV_L	CGOUT20_L[4:0]
186			14		-	CGO21_PH_L	CGO21_INV_L	CGOUT21_L[4:0]
187			15		-	CGO22_PH_L	CGO22_INV_L	CGOUT22_L[4:0]
188	SET_GI P_R	P2	16	1	NVALID	CGO1_PH_R	CGO1_INV_R	CGOUT1_R[4:0]
189			17		-	CGO2_PH_R	CGO2_INV_R	CGOUT2_R[4:0]
18A			18		-	CGO3_PH_R	CGO3_INV_R	CGOUT3_R[4:0]
18B			19		-	CGO4_PH_R	CGO4_INV_R	CGOUT4_R[4:0]
18C			1A		-	CGO5_PH_R	CGO5_INV_R	CGOUT5_R[4:0]
18D			1B		-	CGO6_PH_R	CGO6_INV_R	CGOUT6_R[4:0]
18E			1C		-	CGO7_PH_R	CGO7_INV_R	CGOUT7_R[4:0]
18F			1D		-	CGO8_PH_R	CGO8_INV_R	CGOUT8_R[4:0]
190			1E		-	CGO9_PH_R	CGO9_INV_R	CGOUT9_R[4:0]
191			1F		-	CGO10_PH_R	CGO10_INV_R	CGOUT10_R[4:0]
192			20		-	CGO11_PH_R	CGO11_INV_R	CGOUT11_R[4:0]
193			21		-	CGO12_PH_R	CGO12_INV_R	CGOUT12_R[4:0]
194			22		-	CGO13_PH_R	CGO13_INV_R	CGOUT13_R[4:0]
195			23		-	CGO14_PH_R	CGO14_INV_R	CGOUT14_R[4:0]
196			24		-	CGO15_PH_R	CGO15_INV_R	CGOUT15_R[4:0]
197			25		-	CGO16_PH_R	CGO16_INV_R	CGOUT16_R[4:0]
198			26		-	CGO17_PH_R	CGO17_INV_R	CGOUT17_R[4:0]
199			27		-	CGO18_PH_R	CGO18_INV_R	CGOUT18_R[4:0]
19A			28		-	CGO19_PH_R	CGO19_INV_R	CGOUT19_R[4:0]
19B			29		-	CGO20_PH_R	CGO20_INV_R	CGOUT20_R[4:0]
19C			2A		-	CGO21_PH_R	CGO21_INV_R	CGOUT21_R[4:0]
19D			2B		-	CGO22_PH_R	CGO22_INV_R	CGOUT22_R[4:0]
19E	SET_GI P_L_GS	P2	2C	1	NVALID	-	-	CGOUT1_L_GS[4:0]
19F			2D		-	-	-	CGOUT2_L_GS[4:0]
1A0			2E		-	-	-	CGOUT3_L_GS[4:0]
1A1			2F		-	-	-	CGOUT4_L_GS[4:0]
1A2			30		-	-	-	CGOUT5_L_GS[4:0]
1A3			31		-	-	-	CGOUT6_L_GS[4:0]
1A4			32		-	-	-	CGOUT7_L_GS[4:0]
1A5			33		-	-	-	CGOUT8_L_GS[4:0]
1A6			34		-	-	-	CGOUT9_L_GS[4:0]
1A7			35		-	-	-	CGOUT10_L_GS[4:0]
1A8			36		-	-	-	CGOUT11_L_GS[4:0]
1A9			37		-	-	-	CGOUT12_L_GS[4:0]

1AA			38		-	-	-	CGOUT13_L_GS[4:0]
1AB			39		-	-	-	CGOUT14_L_GS[4:0]
1AC			3A		-	-	-	CGOUT15_L_GS[4:0]
1AD			3B		-	-	-	CGOUT16_L_GS[4:0]
1AE			3C		-	-	-	CGOUT17_L_GS[4:0]
1AF			3D		-	-	-	CGOUT18_L_GS[4:0]
1B0			3E		-	-	-	CGOUT19_L_GS[4:0]
1B1			3F					CGOUT20_L_GS[4:0]
1B2			40					CGOUT21_L_GS[4:0]
1B3			41		-	-	-	CGOUT22_L_GS[4:0]

1B4	SET_GIP_ R_GS	P2	42	1	NVALID	-	-	CGOUT1_R_GS[4:0]				
1B5			43		-	-	-	CGOUT2_R_GS[4:0]				
1B6			44		-	-	-	CGOUT3_R_GS[4:0]				
1B7			45		-	-	-	CGOUT4_R_GS[4:0]				
1B8			46		-	-	-	CGOUT5_R_GS[4:0]				
1B9			47		-	-	-	CGOUT6_R_GS[4:0]				
1BA			48		-	-	-	CGOUT7_R_GS[4:0]				
1BB			49		-	-	-	CGOUT8_R_GS[4:0]				
1BC			4A		-	-	-	CGOUT9_R_GS[4:0]				
1BD			4B		-	-	-	CGOUT10_R_GS[4:0]				
1BE			4C		-	-	-	CGOUT11_R_GS[4:0]				
1BF			4D		-	-	-	CGOUT12_R_GS[4:0]				
1C0			4E		-	-	-	CGOUT13_R_GS[4:0]				
1C1			4F		-	-	-	CGOUT14_R_GS[4:0]				
1C2			50		-	-	-	CGOUT15_R_GS[4:0]				
1C3			51		-	-	-	CGOUT16_R_GS[4:0]				
1C4			52		-	-	-	CGOUT17_R_GS[4:0]				
1C5			53		-	-	-	CGOUT18_R_GS[4:0]				
1C6			54		-	-	-	CGOUT19_R_GS[4:0]				
1C7			55		-	-	-	CGOUT20_R_GS[4:0]				
1C8			56		-	-	-	CGOUT21_R_GS[4:0]				
1C9			57		-	-	-	CGOUT22_R_GS[4:0]				
1CA	SETGIP1	P2	58	1	NVALID	GIP_GAS _OPT	TGEQL	GIP_RST _EN	INIT_PORCH[3:0]			
1CB			59		INIT_W[3:0]				-	INIT[10:8]		
1CC			5A		INIT[7:0]							
1CD			5B		-	-	STV_NUM[1:0]		-	-	-	-
1CE			5C		STV_S0[7:0]							
1CF			5D		STV_W[3:0]				-	STV_S1[2:0]		
1D0			5E		-	-	-	STV_S2[4:0]				
1D1			5F		-	-	-	STV_S3[4:0]				
1D2			60		ETV_W[3:0]				-	ETV_S1[2:0]		
1D3			61		-	-	-	ETV_S2[4:0]				
1D4			62		-	-	-	ETV_S3[4:0]				
1D5			63		SETV_ON[7:0]							
1D6			64		SETV_OFF[7:0]							
1D7			65			ETV_EN	ETV_NUM[1:0]		-	ETV_S0[10:8]		
1D8			66		ETV_S0[7:0]							
1D9			67		CKV0_NUM[3:0]				CKV0_W[3:0]			

1DA			68		CKV0_S0[7:0]								
1DB			69		CKV0_ON[7:0]								
1DC			6A		CKV0_OFF[7:0]								
1DD			6B		CKV0_DUM[7:0]								
1DE	SETGIP2	P2	6C	1	NVALID	EOLR	FLM_MODE	GEQ_LINE	GEQ_W[3:0]				
1DF			6D		-	-	GEQ_GGND1[5:0]						
1E0			6E		-	-	GEQ_GGND2[5:0]						
1E1			6F		GIPDR[1:0]		VGHO_S EL	VGLO_S EL	VGLO_S EL2	CKV_GR OUP	CKV1_C ON	CKV0_C ON	
1E2			70		CKV1_NUM[3:0]					CKV1_W[3:0]			
1E3			71		CKV1_S0[7:0]								
1E4			72		CKV1_ON[7:0]								
1E5			73		CKV1_OFF[7:0]								
1E6			74		CKV1_DUM[7:0]								
1E7			75		FLM_EN	FLM_W[6:0]							
1E8			76		FLM_ON[7:0]								
1E9			77		VEN_EN	VEN_W[10:8]				FLM_NUM	FLM_OFF[10:8]		
1EA			78		FLM_OFF[7:0]								
1EB			79		VEN_W[7:0]								
1EC			7A		VEN_S0[7:0]								
1ED			7B		VEN_S1[7:0]								
1EE			7C		VEN_DUM[7:0]								
1EF			7D		VEN_ON[7:0]								
1F0			7E		VEN_OFF[7:0]								
1F1			SET_GIP_		P2	7F	1	NVALID	GIP_OPT0[6:0]				
1F2	OPT	80	GIP_OPT1[7:0]										
203	SETGIP3	P2	8E	1	NVALID	-	-	-	-	-	FLM_ON_ LRN_EN	FLM_OFF_ _LRN_EN	
204			8F		AGON_E N	AGON_O PT	AGON_LI NE0	AGON_LI NE1	AGON_DLY[3:0]				
205			90		AGON_S[3:0]					AGON_W[3:0]			
206			91		AGON_ON0[7:0]								
207			92		AGON_ON1[7:0]								

3.2. OTP Programming Flow



3.3. Programming Sequence

Step	Operation
1	Power on and HWRST the module. Then do SLPOOUT.
2	Set 0xE1h = 0x93h, 0xE2h = 0x65h, 0xE3h = 0xF8h to access extension commands.
3	Set page1 CMD(RE0h=0x01h) , then set Page1_R1Dh = 0x01h, Page1_R1Eh = 0x03h(VGH=7.6V) and CMD RECh, OTP_INT_VPP = 1 for using internal power mode. Or using the external power 7.5V to VPP.
4	Wait 100ms for VPP power stable.
5	Write optimized OTP value to related register.
6	Set Page0 CMD(RE0h=0x00h), then set Page0_0x84h = 0x5Ah, Page0_0x85h=0xA5h to enable UP KEY.
7	Set 0xEAh 0xEBh, UP_INDEX[10:0] that you want to programmed.
8	Set 0xECh, OTP_AUTO_PROG=1.
9	Delay 100ms for internal power to program OTP. Or delay 10ms for external power to program OTP.
10	One OTP block programming was completed. Return to step (7) for next OTP block programming. Or go next step if all OTP programming is finished.
11	Set Page0_0x84h = 0x00h, Page0_0x85h=0x00h to disable OTP KEY.
12	If you use external power, remove the external power 7.5V from VPP.

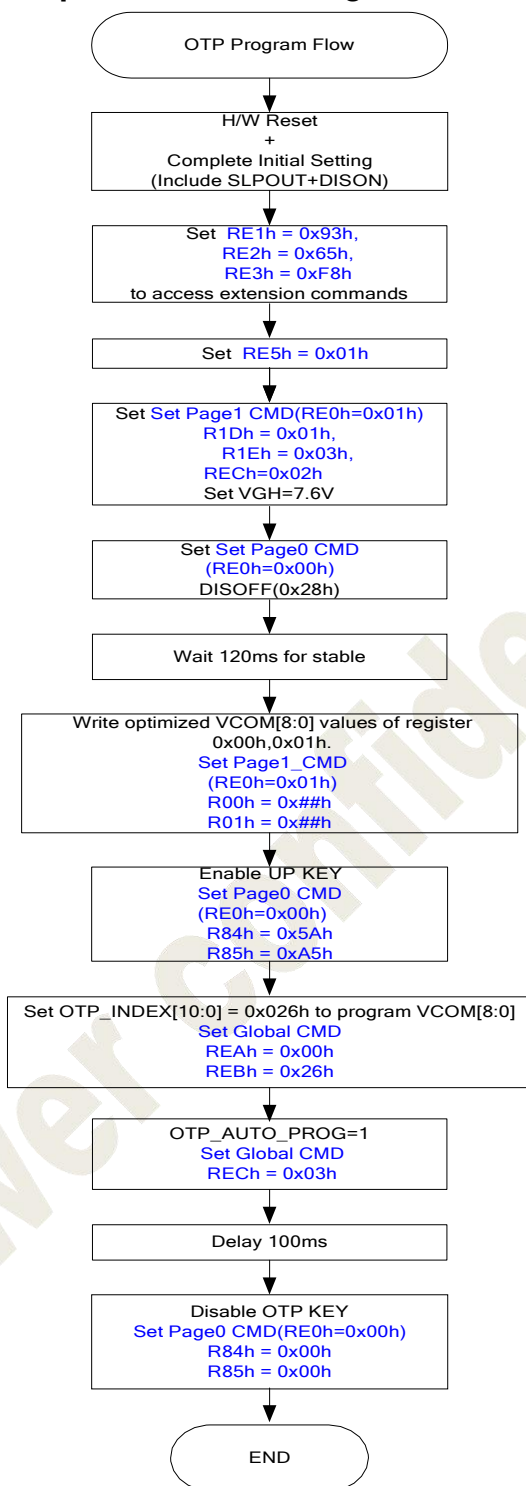
Note:

1. All OTP default value is 1.
2. OTP can be reloaded only NVALID bit is burned(value is 0).
3. Fifipower driver IC do auto OTP programming. User only set the start OTP_INDEX of the Ref. CMD block then all block is programmed by driver IC. This function also can be used for MTP Ref. CMD block.

For example:

- a. To program VCOM_SET 1st time, user set OTP_INDEX=026h then from OTP_INDEX=026h to OTP_INDEX=027h(Ref. CMD block of 1st VCOM_SET) is programmed by driver IC.
- b. To program VCOM_SET 2nd time, user set OTP_INDEX=026h then from OTP_INDEX=028h to OTP_INDEX=029h(Ref. CMD block of 2nd VCOM_SET) is programmed by driver IC.

3.4. OTP Programming example of VCOM setting



3.5. OTP read example

